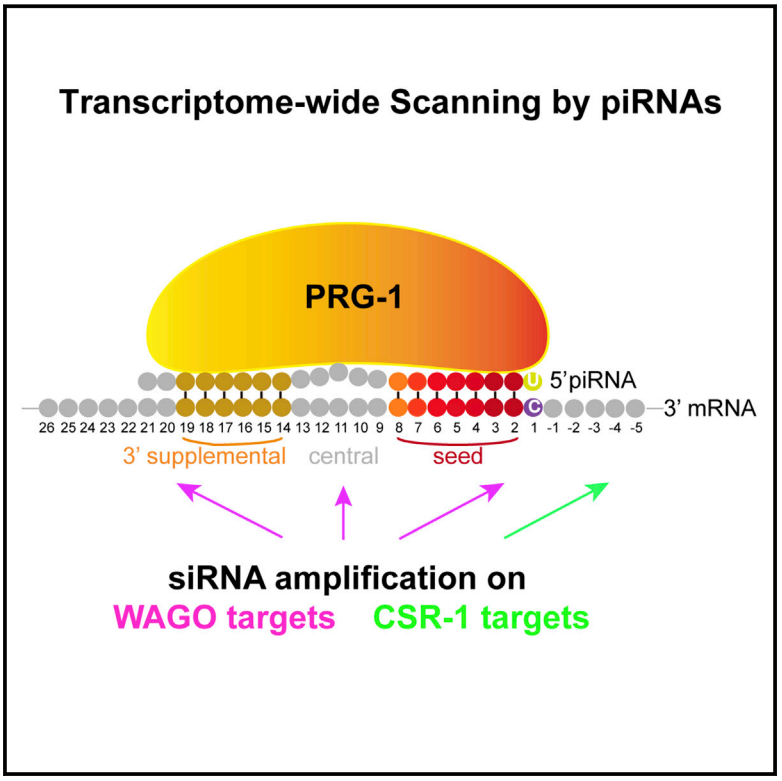
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Identification of piRNA Binding Sites Reveals the Argonaute Regulatory Landscape of the *C. elegans* Germline

Graphical Abstract



Authors

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In Brief

Transcriptome-wide profiling of piRNA targeting rules provides new insights into the interplay between Argonaute pathways and their physiological roles in *C. elegans*

Highlights

- PIWI CLASH identifies piRNA binding sites transcriptome-wide
- piRNAs engage all germline mRNAs via microRNA-like pairing rules
- piRNA target sites have distinct 22G-RNA patterns on silenced and expressed mRNAs
- Targeting by CSR-1 Argonaute correlates with reduced piRNA binding density

Shen et al., 2018, Cell *172*, 937–951
February 22, 2018 © 2018 Elsevier Inc.
https://doi.org/10.1016/j.cell.2018.02.002

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Identification of piRNA Binding Sites Reveals the Argonaute Regulatory Landscape of the *C. elegans* Germline

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SUMMARY

piRNAs (Piwi-interacting small RNAs) engage Piwi Argonautes to silence transposons and promote fertility in animal germlines. Genetic and computational studies have suggested that *C. elegans* piRNAs tolerate mismatched pairing and in principle could target every transcript. Here we employ *in vivo* cross-linking to identify transcriptome-wide interactions between piRNAs and target RNAs. We show that piRNAs engage all germline mRNAs and that piRNA binding follows microRNA-like pairing rules. Targeting correlates better with binding energy than with piRNA abundance, suggesting that piRNA concentration does not limit targeting. In mRNAs silenced by piRNAs, secondary small RNAs accumulate at the center and ends of piRNA binding sites. In germline-expressed mRNAs, however, targeting by the CSR-1 Argonaute correlates with reduced piRNA binding density and suppression of piRNA-associated secondary small RNAs. Our findings reveal physiologically important and nuanced regulation of individual piRNA targets and provide evidence for a comprehensive post-transcriptional regulatory step in germline gene expression.

INTRODUCTION

Argonaute (AGO) proteins and their associated small RNAs are fundamental regulators of transcriptional and post-transcriptional gene regulation (Czech and Hannon, 2011; Ghildiyal and Zamore, 2009; Hutvagner and Simard, 2008; Meister, 2013; Siomi and Siomi, 2009; Thomson and Lin, 2009). Piwi proteins are members of the RNaseH-related Argonaute superfamily and associate with small RNAs (i.e., Piwi-interacting RNAs or piRNAs) to form piRNA-induced silencing complexes (piRISCs) (Czech and Hannon, 2016; Malone and Hannon, 2009; Weick and Miska, 2014). The genomic origins, sequences, and lengths of animal piRNAs vary, but some of the biological functions of piRISCs appear to be shared. For example, piRISCs are required for fertility and transposon silencing in worms, flies, and mice (Aravin et al., 2001, 2007; Batista et al., 2008; Carmell et al., 2007; Siomi et al., 2011; Thomson and Lin, 2009). A growing number of studies suggest that piRNAs and Piwi Argonautes may regulate many, if not all, germline mRNAs (Fagegaltier et al., 2016; Vourekas et al., 2016; Zhang et al., 2015b).

The *C. elegans* Piwi protein, PRG-1, binds an abundant class of germline-expressed 21-nucleotide (nt) piRNAs with a 5′ uridine (21U-RNAs)(Batista et al., 2008; Ruby et al., 2006). Targeting by the 21U-RNA/PRG-1 piRISC complex recruits an RNA-dependent RNA polymerase (RdRP) that initiates the *de novo* synthesis of secondary 22-nt small RNAs that are templated directly from the target RNA and exhibit a bias for a 5′ guanosine residue. These so-called 22G-RNAs engage an expanded group of worm Argonautes (WAGOs) that function downstream of piRNAs to silence transposons and many endogenous genes. The WAGO pathway is required for long-term maintenance of silencing (Bagijn et al., 2012; Lee et al., 2012).

piRNA targeting in *C. elegans* permits mismatches, suggesting that thousands of endogenous mRNAs could be targeted by piRNAs (Lee et al., 2012). However, anti-silencing mechanisms are thought to prevent or reduce the sensitivity of endogenous mRNAs to piRNA-mediated silencing. The CSR-1 pathway, for example, is thought to be one arm of a "self" recognition pathway that protects endogenous mRNAs from piRNA surveillance (Seth et al., 2013). CSR-1 engages RdRP-derived small RNAs templated from nearly all germline-expressed mRNAs (Claycomb et al., 2009). However, it is unknown whether CSR-1 blocks PRG-1 targeting directly or if it acts downstream to prevent WAGO recruitment.

The identification of piRNA targets is essential for deciphering the roles of piRNAs in both sequence-directed immunity and more broadly in the regulation of germline gene expression. Here, we optimize a crosslinking, ligation, and sequencing of hybrids (CLASH) protocol to identify piRNAs and associated

(candidate) target RNA binding sites in *C. elegans* (Helwak et al., 2013; Van Nostrand et al., 2016; Vourekas and Mourelatos, 2014). We identified 200,000 high-confidence piRNA-target site interactions. The overwhelming majority of interactions were between piRNAs and mRNAs. Bioinformatics analysis of the hybrids revealed that targets are enriched for energetically favorable Watson-Crick pairing with their associated piRNAs. We show that the seed sequence (i.e., positions 2 to 8) and supplemental nucleotides near the 3′ end (positions 14 to 19) of the piRNA are important determinants of piRNA-target binding and silencing, suggesting that piRNA targeting resembles miRNA targeting.

piRNA target sites defined by CLASH show a non-random pattern of WAGO 22G-RNAs that initiate at both ends and near the center (position 12) of the piRNA target site, consistent with local recruitment of RdRP. Analysis of CLASH hybrids obtained from CSR-1-depleted animals suggest that CSR-1 protects its targets from PRG-1 binding and WAGO-dependent silencing. Our findings reveal that the entire germline mRNA transcriptome engages piRISC, and suggests how germline Argonaute pathways are coordinated to achieve comprehensive regulation and surveillance of germline gene expression.

RESULTS

PRG-1 CLASH Directly Identifies piRNA-Target Chimeras

We used a modified cross-linking, ligation, and sequencing of hybrids (CLASH) approach to identify RNAs associated with the *C. elegans* PRG-1-piRISC complex. Briefly, CLASH involves the *in vivo* cross-linking of RNAs to a protein of interest followed by immunoprecipitation (IP), trimming of RNA ends, ligation to form hybrids between proximal RNAs within the crosslinked complex, cDNA preparation, library construction, and deep sequencing (see STAR Methods; Figures 1A–1E). In principle, this procedure should allow the recovery of hybrid-sequence reads formed when piRNAs are ligated to proximal cellular target RNAs within the cross-linked PRG-1 IP complex.

To perform CLASH, we first used CRISPR/Cas9 genome editing to introduce a GFP-TEV-FLAG (GTF) multiplex tag into the endogenous *prg-1* locus (Kim et al., 2014). In addition to direct fluorescence detection this tag also permits tandem affinity purification with a TEV protease-mediated elution after the first affinity step. GTF::PRG-1 exhibited a robust expression and was prominently localized in P-granules (Figure S1A) in a pattern identical to that previously reported in PRG-1 immunolocalization studies (Batista et al., 2008). Moreover, the GTF::PRG-1 fusion protein was functional, as evidenced by its ability to mediate piRNA-dependent silencing of a *gfp::cdk-1* reporter gene (Figure S1B).

We then carefully optimized each step of the CLASH procedure using GTF-PRG-1 (Figures 1A–1E; STAR Methods) (Broughton et al., 2016; Helwak et al., 2013). The tandem affinity purification of GTF::PRG-1 resulted in recovery of a single prominent protein of the expected size in silver-stained SDS-PAGE gels (Figure 1B). Although GTF::PRG-1 stably associated with piRNAs under these purification conditions, the recovery of longer associated RNA required the pretreatment of the worms with ultraviolet light. These crosslinked RNA Protein complexes, RNPs, were then treated with nuclease to trim the long RNAs, followed by intermolecular ligation to form RNA hybrids between the piRNA and target (Figures 1B–1D). RNA hybrids of approximately 42 nts were recovered by gel purification (Figure 1E) and were used for library construction and deep-sequencing.

In two independent experiments, we found similar distributions of mapped sequence reads (Figures S1C–S1H). Together, these comprised a total of ~21million reads, including a total of ~7-million reads corresponding to 17,192 different piRNAs. Most of these piRNA-containing reads lacked a hybrid sequence (1,083,172), or the hybrid sequences could not be mapped to the genome because they were too short, or for other reasons (3,946,162). We obtained 2,106,813 hybrid reads composed of a piRNA sequence and a genome-mapping sequence, of which ~1.5 million were composed of a single piRNA sequence fused to an mRNA. In addition to mRNA chimeras, we detected piRNAs fused to sequences corresponding to rRNA (137,322 reads), tRNA (11,231 reads), pseudogenes (48,208 reads), lincRNA (2,583 reads), miRNA (1,556 reads), introns (10,529 reads), and transposable elements (19,092 reads).

CLASH Reveals piRNA Target Sites in Germline mRNAs

Because mRNA chimeras were by far the most abundant type of hybrid read, we chose to focus on mRNA hybrids in the present study. Altogether a total of 16,385 genes were represented among the piRNA hybrids (Figure 1F). We found that "soma-specific" mRNAs were strongly under-represented in the CLASH data (Figure 1G) (Beanan and Strome, 1992; Li et al., 2014), consistent with the idea that CLASH captures interactions between piRNAs and mRNAs that occur in the germline, and not interactions that occur in lysates. The frequency of recovery of each piRNA by CLASH correlated with its level in the input sample as measured by small RNA sequencing (Figure S1I, $r = 0.58$, $p < 0.005$).

The nuclease treatment during the CLASH procedure was optimized to produce chimeras of approximately 40 nucleotides. Thus, each chimera potentially reflects a piRNA/target mRNA duplex ligated at, or near, one end of the duplex. We noted, however, that not all chimeras contained a full-length piRNA and that the recovered target regions varied in length, indicating some variability in nuclease trimming during the CLASH procedure. Therefore, prior to searching for base-pairing interactions, we inferred the full-length piRNA and extended the empirically defined target space by adding nucleotides to each end, creating "ideal" piRNA/target RNA pairs (see STAR Methods).

We next predicted the most energetically favorable piRNA-mRNA interactions from *in silico* folding of these "ideal" sequences and compared it with predicted binding energies in a control dataset with randomly matched pairs (Figure 1H). This analysis showed that stable base-pair interactions were strongly enriched in the recovered piRNA-mRNAs chimeras. In fact, when normalized for mRNA levels, hybrid read counts per target site correlated better with binding energy than with piRNA abundance (see Discussion) (Figure 1I). Chimeras in which the piRNA 3′ end was contiguous with mRNA sequence were roughly 20-fold more frequent than chimeras ligated at piRNA 5′ ends

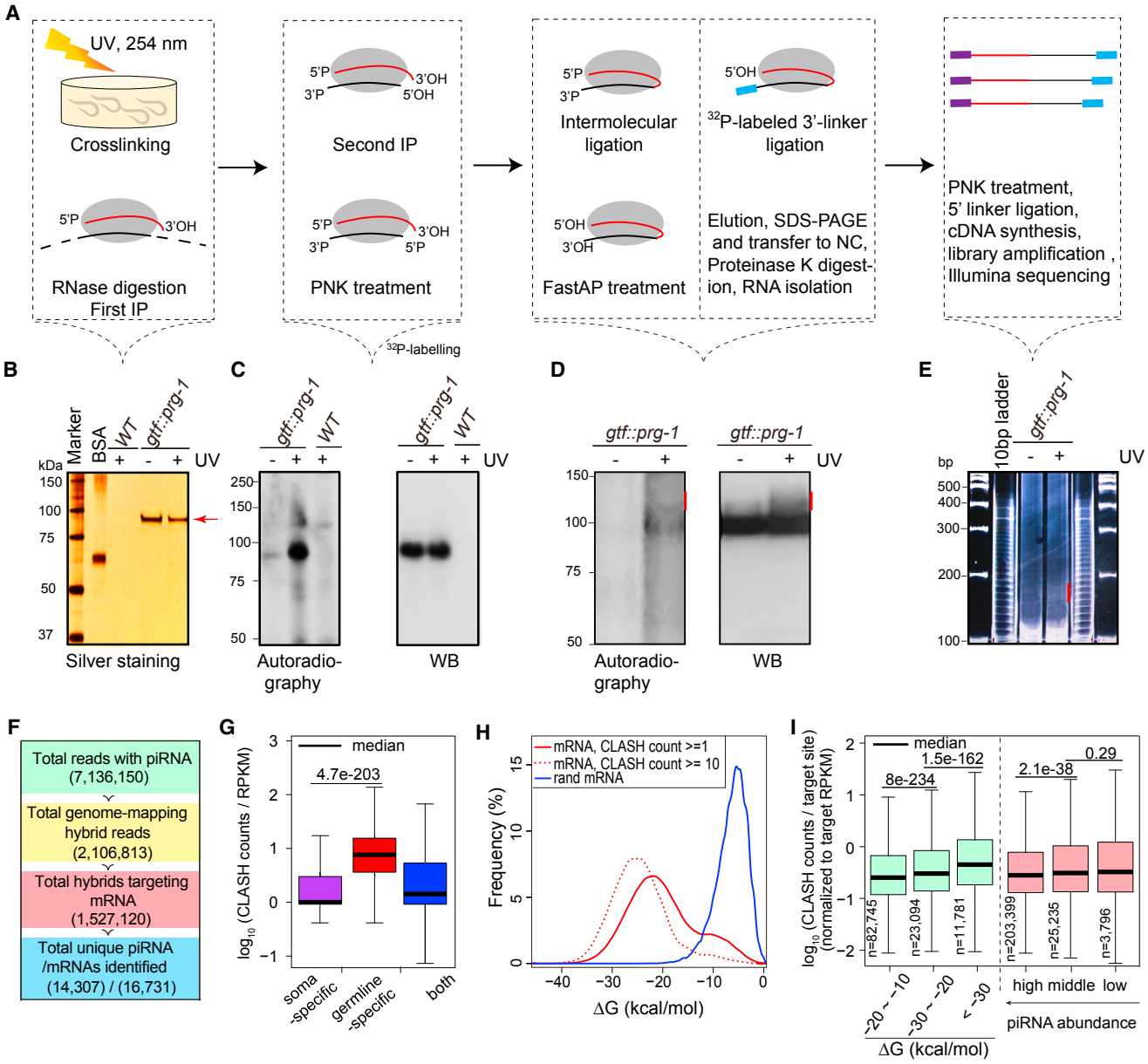


Figure 1. PRG-1 CLASH Identifies piRNA-Target Chimeras in *C. elegans*

(A) CLASH workflow. PRG-1 (gray oval), piRNA (red line), and mRNA (black line). Linkers indicated as blue (3′) and purple (5′) rectangles.

(B) Silver stain analysis of GFP IP complexes (red arrow) purified from wild-type (WT) and *gtf::prg-1* worms with or without UV irradiation.

(C and D) Autoradiography (left) and western blot (right) of FLAG IP complexes released by TEV cleavage after GFP IP. Panels C and D show independent samples. Red line shows the region excised for library preparation.

(E) Polyacrylamide gel showing library products (red line) isolated for sequencing.

(F) Summary of combined data from two CLASH replicates.

(G) Normalized CLASH counts per gene for soma-specific genes, germline-specific genes, and both. CLASH reads normalized as described in STAR Methods.

(H) Predicted binding energies (ΔG, kcal/mol) between piRNAs and target sites identified by CLASH (red) or between randomly matched pairs (blue).

(I) Boxplots of CLASH counts per target site with increasing binding energy or piRNA abundance (low, 0%–33%; medium, 33%–66%; high, 66%–100%). Median, solid black line. Significant differences between groups indicated by p values.

See also Figure S1.

(Figure S1J). These findings are consistent with the idea that piRNA 3′ ends are more available for ligation to their targets. Taken together, these findings support the idea that CLASH captures proximal mRNAs bound to piRISC via base-pairing interactions.

piRNA Targets Exhibit a Pattern of Discrete Peaks in 22G-RNA Levels

In *C. elegans*, piRISC recruits RdRP to its targets. Therefore, we wished to examine the pattern of RdRP-dependent 22G-RNA

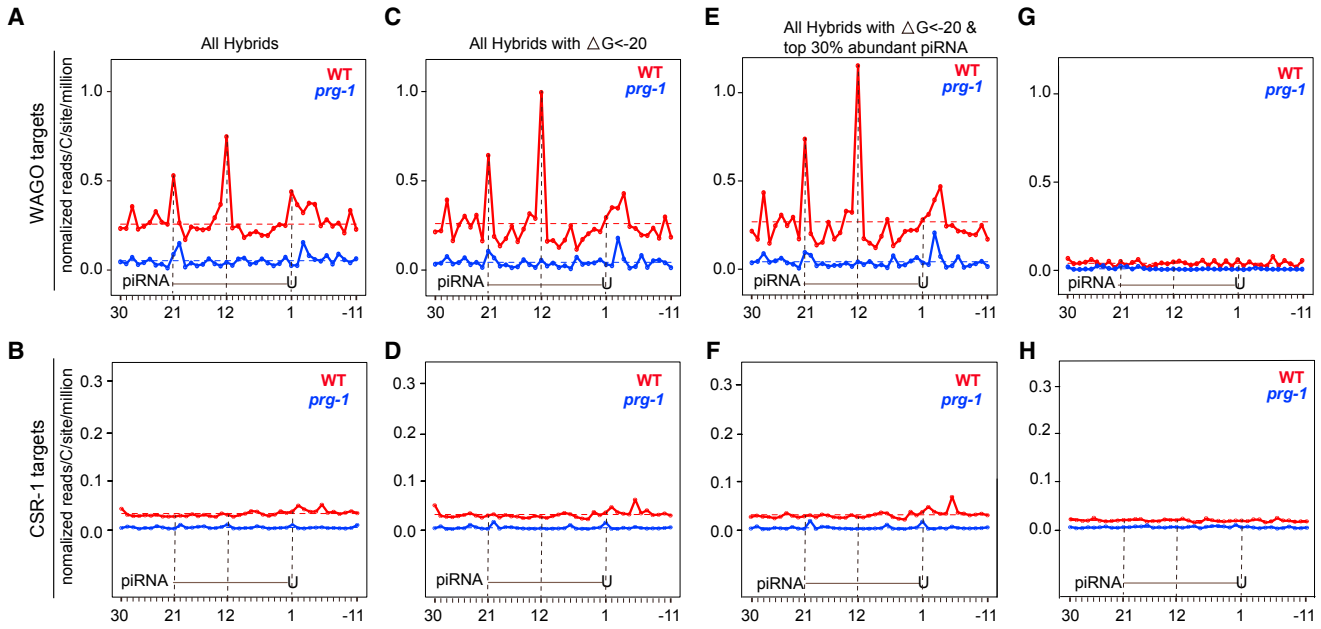


Figure 2. 22G-RNAs Peak at the Center and Ends of piRNA Binding Sites

(A–H) 22G-RNA 5′ ends cloned from wild-type (red) or *prg-1* (blue) worms mapped at single-nucleotide resolution to an extended 40-nt window around CLASH-defined piRNA target sites. Each plot is centered on a 21-nt piRNA, shown schematically. WAGO (A, C, E) and CSR-1 (B, D, F) targets analyzed separately. All hybrids (A and B), hybrids with ΔG < –20 kcal/mol (C and D), hybrids from the high abundance piRNAs with ΔG < –20 kcal/mol (E and F), and control target regions > 100-nt from the defined piRNA target sites (G and H).

See also Figure S2.

production near CLASH-defined piRNA target sites in both WT and *prg-1* mutant worms. To do this, we plotted 22G-RNA levels within a 40-nt region centered on the piRNA complementary sequences defined by CLASH. The 5′ ends of 22G-RNAs are thought to be formed directly from RdRP initiating at C residues within the target mRNAs. We therefore normalized the 22G-RNA levels initiating at each position to the frequency of C residues within the CLASH-defined targets at each position. Because the CSR-1, and WAGO Argonaute pathway are thought to have opposing functions, resisting and supporting piRNA silencing (Seth et al., 2013; Wedeles et al., 2013), we separately considered predicted piRNA targets within previously defined WAGO and CSR-1 targeted mRNAs (Claycomb et al., 2009; Gu et al., 2009). As a control set, we considered a target region arbitrarily set 100 nts away (within each mRNA) from of the piRNA binding sites identified by CLASH. In WT animals, 22G-RNA levels were much higher for WAGO targets than for CSR-1 targets, as expected (Figures 2A and 2B, Figures S2A and S2B). However, piRNA binding sites within both WAGO and CSR-1 targets showed a non-random distribution of 22G-RNA levels across the interval. By contrast, the control regions within the same target mRNAs, but offset from the hybrid sites, exhibited no such patterns (Figures 2G and 2H). WAGO targets exhibited a strong central peak, and clusters of peaks at either end of the piRNA target sites. To describe these patterns, we refer to the mRNA sequences near the target site as follows: t1 through t30 includes the presumptive binding site (t1 to t21) plus 9 nucleotides 5′ of the target site (t22 to t30). The mRNA region 3′ of the target site consists of nucleotides t–1 through t–11. Strikingly, this analysis revealed a prominent peak in the center of the piRNA complementary region near t12, and smaller peaks centered at t1 and t21 (Figure 2A). CSR-1 targets exhibited a cluster of much smaller peaks near the 3′ end of the predicted target site, with the largest peak residing in sequences located near t–5 (Figure 2B). The amplitudes of 22G-RNA levels on both the WAGO and CSR-1 targets correlated positively with the predicted free energy of piRNA binding and to a lesser extent with piRNA abundance (Figures 2C–2F and S2C–2F).

The amplitude and position of 22G-RNA peaks differed in *prg-1* mutants. For WAGO targets, the central peak at t12 was completely depleted in *prg-1* mutants, whereas the terminal peaks were reduced. In CSR-1 targets, the prominent peak located at t–5 disappeared, but new peaks at t1, t6, and t21 became evident (Figure 2F). This analysis suggests that PRG-1 influences both the precise position, and the levels of 22G-RNAs on its targets, and that CSR-1 and WAGO targets differ strikingly in their accumulation of 22G-RNAs in response to piRNA targeting (see Discussion).

Patterns of piRNA Targeting

Previous studies have revealed features of Argonaute/small RNA guided targeting, including the importance of ''seed'' pairing between the target and nucleotides 2 to 8 of the small RNA guide (Bartel, 2009). To explore patterns of piRNA-mediated targeting we used two independent computational strategies. In the first strategy, we considered the *in silico* predicted folding within a

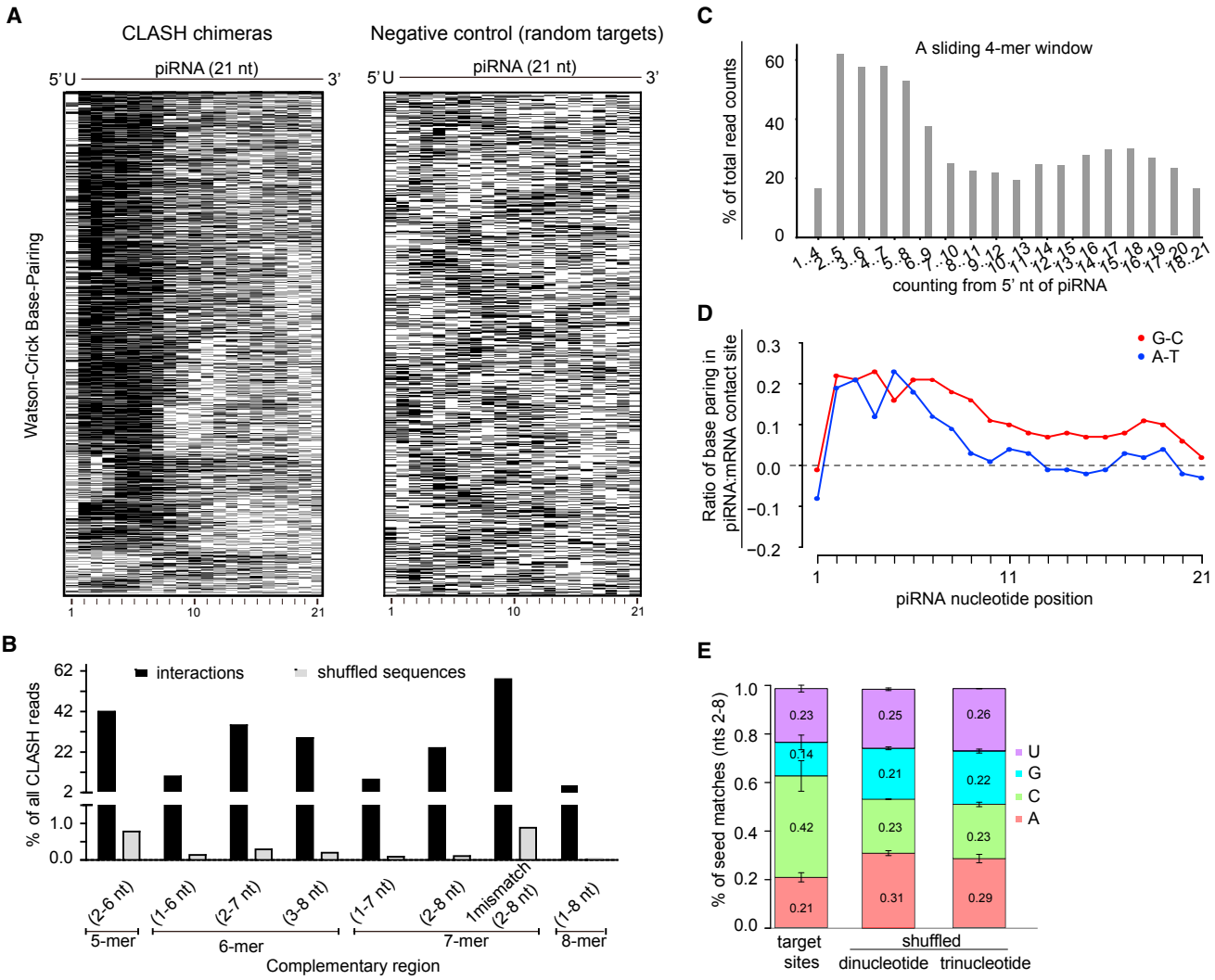


Figure 3. piRNAs Target with miRNA-Like Seed and 3′ Supplementary Base Pairing

(A) Heatmaps of Watson-Crick base-pairing (black pixels) at each piRNA position for piRNA-mRNA chimeras detected at least 5 times (left), and for negative control (random target) sequences.

(B) Chart showing the percentage of CLASH reads (piRNA-target chimeras; black) with complementarity at the indicated positions. Approximately 70% of piRNA-target interactions possess the tested complementarities. Target sequences with shuffled dinucleotides served as control (gray).

(C) A 4-mer sliding window search for perfect Watson-Crick base pairing between piRNAs and CLASH-defined targets.

(D) Ratios of G:C (red) or A:T (blue) base pairing in piRNA-mRNA duplexes, after deducting ratios from random control.

(E) Percent of seed-matched target sites (left) with the indicated nucleotide at target position 1 (t1; opposite piRNA position 1). Randomized target sequences with shuffled dinucleotides (center) and trinucleotides (right) serve as controls. Data expressed as mean ± 2 s.e.m. from two replicates.

See also Figure S3.

high-confidence group of "ideal" piRNA/target RNA pairs that were identified by at least 5 sequence reads in our combined datasets. To identify preferred base-pairing patterns within this group of hybrids, we applied the Affinity Propagation clustering algorithm (APcluster) (Frey and Dueck, 2007). This analysis revealed a clearly preferred interaction at the seed region and distinct base-pairing patterns at the 3′ supplementary region (Figures 3A and 3B, and Figure S3A). Notably, base-pairing frequencies declined from positions 9 to 13 of the piRNA and increased from positions 14 to 19 (Figure S3B). As expected, these patterns were not enriched in a set of randomized piRNA target RNA pairs (Figure 3A, and Figure S3A).

In the second approach, we performed base-pairing analysis using a sliding 4-, 5-, or 7-nt window of piRNA sequence to search for Watson-Crick pairing in each RNA target (Figures 3C, and Figure S3C). Consistent with the analysis in Figures 3A and 3B, this approach revealed a pattern of seed and supplementary pairing, with a distinct drop in pairing from positions 9 to 13 (Figure 3C and Figure S3C). Taken together, these findings suggest that both seed pairing at positions 2 to 8 and

supplementary pairing at positions 14 to 19 contribute to piRNA-target RNA binding (Shin et al., 2010; Wee et al., 2012).

To further characterize piRNA-mRNA interactions, we analyzed A:U and G:C base-pair ratios at each position of the piRNA. We found no significant difference between the two base pair ratios within the seed region, but in other regions, we found a bias toward G:C pairing (Figure 3D). Notably, cytosine was strongly over-represented in the target strand immediately 3′ of the seed complement opposite the 5′ u, (defined as target strand position 1 cytosine, or "t1C") (Figure 3E). This preference contrasts with t1A preferred by insect Piwis (Wang et al., 2014). A search for evolutionary conservation using PhyloP scores (Pollard et al., 2010) failed to reveal a preferential conservation for piRNA-mRNA target sites (Figure S3D).

Finally, we compared the features of piRNA target interactions on WAGO and CSR-1 targets. The energetics of piRNA targeting, the patterns of seed and supplementary pairing, and the average C content along the target region were no different between these groups (Figure S3F).

Seed and 3′ Supplementary Pairing Are Required for Target Silencing

To determine the importance of base pairing interactions along the length of the piRNA/target mRNA hybrid, we used CRISPR genome editing to systematically mutate positions 2 to 21 of an *anti-gfp* piRNA expressed from the *21ux-1* piRNA locus (Figure 4A, Figures S4A–4F; Seth et al., 2018). We then assayed the ability of each *21ux-1(anti-gfp)* mutant piRNA to silence a single-copy *cdk-1::gfp* transgene over a time course of up to 8 generations (Figures 4B and 4C). Strikingly, we found that individual mismatches in the seed region (i.e., m2 to m8) and 3′ supplemental region (i.e., m14 to m21) strongly reduced the ability of *21ux-1(anti-gfp)* to silence *cdk-1::gfp*, but mismatches at the central region (m9 to m13) had a much more mild effect (Figure 4B). By the F2 generation, when fully matched *21ux-1(anti-gfp)* piRNA silences *cdk-1::gfp* by 70% (Figure S4G), mismatches at positions 2 to 8 or 14 to 21 reduced silencing to less than 10% and 25% (respectively) of animals scored (Figure 4B). By contrast, mismatches at positions 9 to 13 reduced silencing activity only slightly, to approximately 50% at the F2 generation. Mismatches at positions 2 or 3 prevented silencing of *cdk-1::gfp*, even after 8 generations, demonstrating that pairing at positions 2 and 3 is essential for piRNA-mediated silencing. Mutants with mismatches at any of the other 18 positions eventually silenced *cdk-1::gfp* over the 8 generation time course (Figure 4C, Figure S4H). Consistent with these findings, we also observed by western blotting after generations 4 and 8 that mutants with mismatches at positions 2 to 8 or 14 to 21 produced much higher levels of CDK-1::GFP protein than did mutants with mismatches at other positions (Figure 4D).

To further test the importance of pairing in these regions, we selectively mutated positions t3, t15, and t21 of the *anti-gfp* target site in *cdk-1::gfp* mRNA to compensate for *anti-gfp* piRNA mutations in guide-strand positions, g3, g15, and g21, each of which strongly diminished silencing. As expected, in the absence of *21ux-1(anti-gfp)*, these silent mutations did not affect the level of GFP expression (Figures S4I–S4K). Strikingly, target mRNAs with "re-matching" mutations at t3, t15, and t21 were each rapidly silenced by piRNA strains bearing the corresponding guide mutations (Figures S4K–S4M). Thus the failure of the g3, g15, and g21 point mutant piRNAs to silence wild-type *cdk-1::gfp* was caused specifically by the mismatches and not by changes in expression or piRISC loading of the mutant piRNAs.

Lastly, we analyzed 22G-RNA induction for several *21ux-1(anti-gfp)* point mutant strains. As expected, we found that 22G-RNA levels correlated with the degree of GFP silencing observed (Figures 4E and 4F). Overall, these findings confirm the importance of base-pairing within the seed region (nucleotides 2 to 8) and within the 3′ supplemental pairing region (nucleotides 14 to 21) for efficient piRNA targeting.

Specific piRNA-mRNA Interactions Suppress Endogenous mRNA Targets

To investigate how the base-pairing rules defined by our bioinformatics and transgene studies affect targeting of an endogenous mRNA, we edited the *21ux-1* target site, introducing single mismatches into the predicted *21ux-1/xol-1* target duplex (Figure 5A). XOL-1 is a key regulator of dosage compensation and sex determination in early zygotes, and *xol-1* mRNA was recently shown to be regulated by the X chromosome-derived piRNA, *21ux-1* (Tang et al., 2018). Consistent with our findings in the transgene studies, single-nucleotide mismatches within the seed and 3′ supplemental pairing regions, but not within the central region, dramatically increased expression of XOL-1 (Figure 5B). The *21ux-1* mutants with mismatches in the seed and 3′ supplemental pairing regions were phenotypically similar to *21ux-1* null mutants and enhanced the dosage compensation and sex determination phenotypes (decreased brood size and masculinization of hermaphrodites) caused by silencing the X-signal element *sex-1* (Figures 5C and 5D) (Carmi et al., 1998). Thus, mutating a single nucleotide in *21ux-1* dramatically increases both XOL-1 expression and activity.

As with most germline mRNAs we found that *xol-1* was targeted by multiple piRNAs. We identified a total of 166 CLASH hybrids containing *xol-1* mRNA sequences fused to 40 different piRNAs (Figure 5E). However, given the importance of *21ux-1* in regulating *xol-1,* and the fact that *21ux-1* is the most abundant piRNA, we were surprised to find that a different piRNA, *21ur-4863*, was recovered in *xol-1* chimeras at a frequency greater than twice that of *21ux-1* chimeras. Specifically, we identified 8 reads with *21ux-1* fused to its *xol-1* target site and 19 reads of *21ur-4863* fused to its *xol-1* target site.

We therefore wished to ask if *21ur-4863* is also important for *xol-1* regulation. Strikingly, deletion of *21ur-4863* resulted in the upregulation of both *xol-1* mRNA and protein levels to a degree similar to that observed in *21ux-1* mutants (Figures 5F and 5G). Similar to the *21ux-1* mutant, the *21ur-4863* deletion mutant enhanced defects in dosage compensation and sex determination caused by *sex-1(RNAi)*: fewer progeny, masculinization of hermaphrodites, embryonic lethality, and dumpy (Dpy) body morphology (Figures 5H and 5I, Figure S5A) (Carmi et al., 1998). Thus, *21u-4863* and *21ux-1* are both required for *xol-1* silencing—neither is sufficient—suggesting that piRNAs function cooperatively to silence *xol-1*.

We also examined the consequences of piRNA targeting on *fbxb-97* and *comt-3*, whose mRNAs are also regulated by

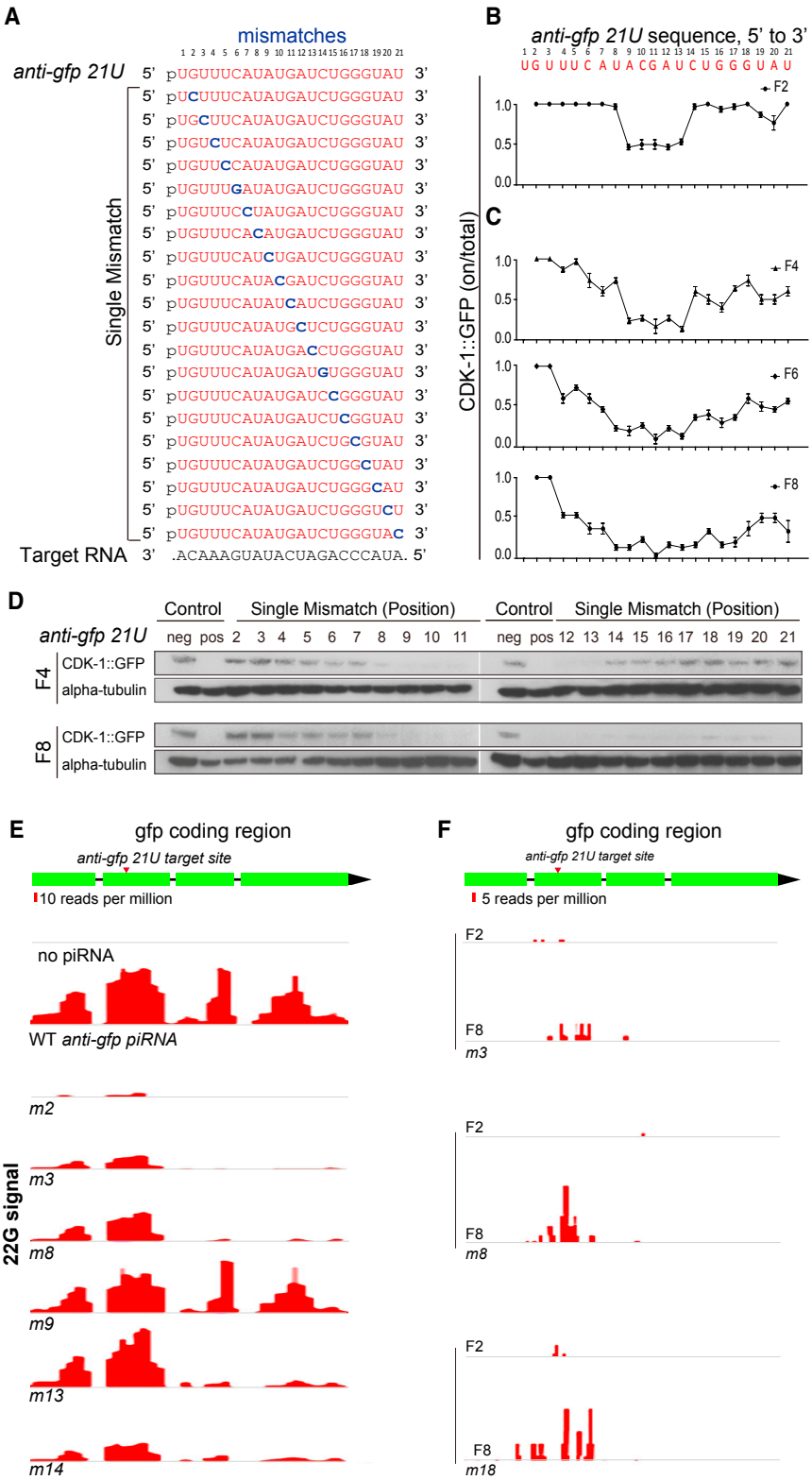


Figure 4. Seed and 3′ Supplementary Pairing Are Required for Silencing

(A) *anti-gfp* piRNA (red) and single-nucleotide mismatches (blue) from positions 2 to 21 on the piRNA target site in *cdk-1::gfp* (black).

(B and C) Graphs of the fraction of GFP-positive worms in the presence of *anti-gfp* piRNA with single-nucleotide mismatches at the indicated piRNA positions (numbers). Ten worms were randomly picked for fluorescence microscopy at the F2 generation (B) and later generations F4, F6, and F8 (C). Data expressed as mean ± 2 s.e.m. of three experiments.

(D) Western blots of CDK-1::GFP in F4 and F8 worms with single-nucleotide mismatches at each piRNA position (from C). Negative control worms without *anti-gfp* piRNA. Positive control worms with fully match *anti-gfp* piRNA.

(E) Schematic of 22G-RNAs targeting *gfp* in F4 *cdk-1::gfp* worms with the indicated single-nucleotide mismatches (m2 = position 2 mismatch, etc.). Positions from 5′, central, and 3′ regions of the piRNA were randomly chosen for analysis. Scale bar, 5 reads per million.

(F) Schematic of 22G-RNAs targeting *gfp* in *cdk-1::gfp* worms with the with the indicated single-nucleotide mismatches (m3, m8, or m18) at the F2 and F8 generations. Scale bar, 5 reads per million. See also Figure S4.

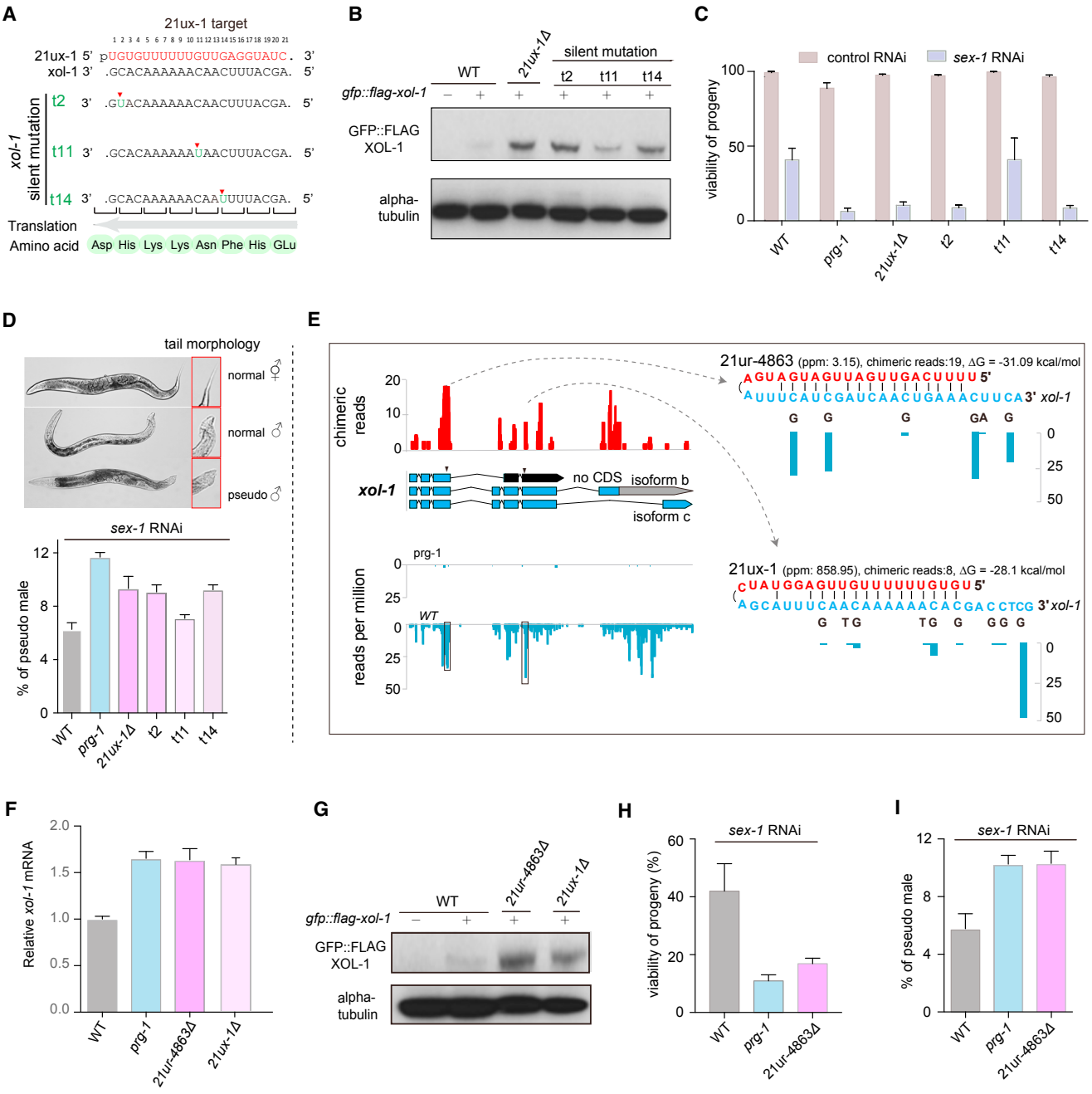


Figure 5. *21ur-4863* and *21ux-1* Suppress *xol-1* Function in Sex Determination

(A) Diagram of silent mutations at positions t2, t11, and t14 of the *21ux-1* target site in *xol-1* (green) to create single-nucleotide mismatches with *21ux-1*.

(B) western blot (anti-FLAG) of GFP::FLAG::XOL-1 levels in *gfp::flag::xol-1* transgenic worms (+) with intact *21ux-1* (WT), *21ux-1* deletion, or *xol-1* silent mutations at position t2, t11, or t14. N2 worms serve as negative control (–).

(C) Bar graphs of percent viable progeny of WT, *prg-1* loss of function, *21ux-1* deletion, or *xol-1* single-nucleotide mismatch (t2, t11, t14) worms treated with *sex-1(RNAi)*. n > 150 progeny per experimental group. Data expressed as mean ± 2 s.e.m. of three experiments.

(D) DIC images (upper panel) of typical hermaphrodite, male, and pseudomale worms. Bar graphs (lower panel) show the percent pseudomale in WT, *prg-1*, *21ux-1* deletion, and *xol-1* single-nucleotide mismatch (t2, t11, t14) worms treated with *sex-1(RNAi)*. n > 100 progeny per experimental group. Data expressed as mean ± 2 s.e.m. of three experiments.

(E) Distribution of chimeric *xol-1* reads (red) identified by CLASH, and the distribution of *xol-1* 22G-RNAs (blue) in *prg-1* mutant and WT worms. Locations of *21ur-4863* (upper) and *21ux-1* target sites in *xol-1* gene indicated by inverted black triangles. Sequences and base pairing (right) of *21ur-4863:xol-1* (upper) and *21ux-1:xol-1* (lower) chimeras. piRNA expression level, number of chimeric reads, and binding energy (ΔG, kcal/mol) indicated above each chimera. Distribution of 22G-RNAs at single-nucleotide resolution shown below each chimera.

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PRG-1 (Bagijn et al., 2012; Batista et al., 2008; Gu et al., 2009; Lee et al., 2012). We identified 70 chimeric reads between *21ur-1563* and *fbxb-97* (Figure S5B). *fbxb-97* mRNA levels were upregulated 1.5-fold in a *21ur-1563* deletion mutant and ~8-fold in the *prg-1* mutant (Figure S5C). To analyze piRNA regulation of *comt-3* (Figure S5D), we took the alternative approach of mutating target sequences. We introduced silent mutations into wobble-positions that maintain the *comt-3* open reading frame but disrupt 4 piRNA target sites (Figure S5E). *comt-3* mRNA levels were markedly increased in the *prg-1* mutant and in the *comt-3* quadruple-piRNA target site mutant, but were not elevated in a *comt-3* single-piRNA target site mutant (Figure S5F). COMT-3::FLAG (introduced by CRISPR) was significantly elevated (by 1.5 fold) in the quadruple target site mutant (Figure S5G). Taken together, our findings suggest that individual piRNAs exhibit a range of regulatory effects and that multiple piRNAs cooperatively silence individual targets.

CLASH Analysis Reveals Competition between the CSR-1 and PRG-1 Argonaute Pathways

Previous studies suggested that CSR-1 protects its germline mRNA targets from piRNA-mediated silencing (Seth et al., 2013; Shirayama et al., 2012; Wedeles et al., 2013). We sought to test whether CSR-1 protects its targets by preventing PRG-1 from binding. To do this, we used an auxin-inducible degradation (AID) system to conditionally deplete CSR-1 in young adult worms (CSR-1^{depleted}; Figure S6A and S6B) (Zhang et al., 2015a), and then performed CLASH on CSR-1^{depleted} worms in two independent biological replicates. We compared the number of unique piRNA binding sites on CSR-1 targets from CSR-1^{depleted} and wild-type worms. Strikingly, we found that the number of unique piRNA binding sites significantly increased (~2 fold) in the CSR-1^{depleted} worms compared to wild-type (Figures 6A and 6B, Figures S6C–S6E). This increase did not result from changes in target mRNA levels, which did not change dramatically during CSR-1 depletion (Figures S6F–S6H). Increased piRNA targeting is illustrated for *dhc-1*, whose mRNA levels did not appreciably change (~1.5 fold), but whose piRNA targeting was elevated by > 3.4-fold in CSR-1^{depleted} worms (Figure 6C). These results suggest that, when CSR-1 is depleted, mRNAs normally targeted by CSR-1 become bound by additional piRNAs.

To determine whether increased piRNA binding correlates with decreased mRNA levels, we plotted the fold change in mRNA abundance (CSR-1^{depleted} / WT) versus the fold change in piRNA-binding density (CSR-1^{depleted} / WT) for 3,820 CSR-1 targets (Figure 6D) (Claycomb et al., 2009). We observed a negative correlation between increased piRNA-binding density and mRNA abundance in CSR-1^{depleted} worms (r = - 0.44). To clearly visualize this relationship, we split the 3,820 CSR-1 targets into five bins of increasing piRNA binding density and plotted the change in mRNA abundance in CSR-1^{depleted} versus wild-type (Figures 6E). This analysis revealed that, as piRNA binding density increases, mRNA abundance decreases. These findings support the idea that CSR-1 functions, at least in part, upstream of PRG-1 to reduce piRNA targeting.

DISCUSSION

In this study, we took the unbiased approach of directly cross-linking piRNAs to target RNAs *in vivo*. The resulting transcriptome-wide snap-shot of piRNA/target-RNA interactions reveals that all germline mRNAs undergo piRNA surveillance. Our findings are consistent with a model for germline gene regulation wherein mRNAs undergo comprehensive post-transcriptional scanning by Argonaute systems. More than 10,000 distinct piRISCs access hundreds of thousands of target sequences on germline mRNAs. Our finding that binding energy was better correlated with hybrid formation than was piRNA abundance, suggests that, for most piRNAs, piRISC concentration is not limiting. Thus surveillance by piRISC is both transcriptome -wide and remarkably efficient. Perhaps as yet unknown features of the enigmatic P-granules, where piRISC resides and presumably functions, create an environment that facilitates this seemingly daunting task of comprehensive mRNA surveillance (Figure 7; see also Seth et al., 2018).

Non-mRNA piRNA Interactions

Although mRNA target sites accounted for greater than 90% of CLASH hybrid reads, we also reproducibly identified CLASH reads mapping to a variety of non-coding RNA species (ncRNAs). For example, over 80,000 CLASH reads and hundreds of different piRNAs were mapped to ncRNA hybrids, including sequences from a single region of the 26S rRNA (Figure S1K). Interestingly, this rRNA region is also targeted by WAGO 22G-RNAs that were recently reported to downregulate rRNA levels in response to stress (Zhou et al., 2017).

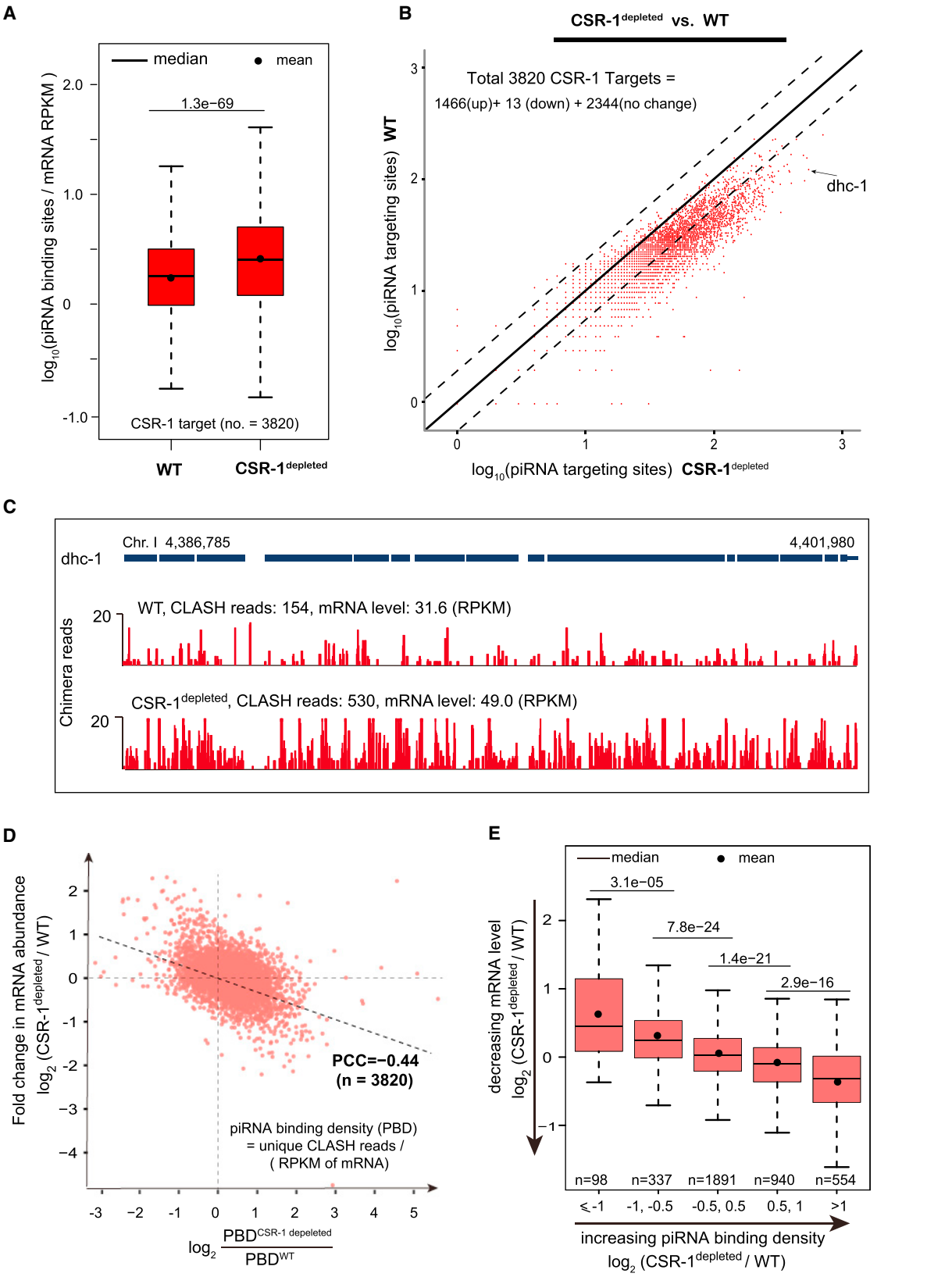
Our studies also identified many interactions between piRNAs and tRNA species. For example, tRNAGlu(CUC) and 21U-8377 formed highly reproducible chimeras that showed thermodynamically stable base-pairing (Figure S1L). Altogether we identified piRNA tRNA hybrids involving 474 different tRNAs and 1225 different piRNAs. The significance of these findings remains to be determined, but it is intriguing that in *Drosophila* a mutation that leads to accumulation of misprocessed tRNA results in a collapse of Piwi-mediated transposon silencing (Molla-Herman et al., 2015; Yamanaka and Siomi, 2015). We also identified hybrids between piRNAs and other ncRNAs including microRNAs and annotated long-noncoding, lncRNAs. The identification of these piRNA interactions provides a new lens through which to

(F) Bar graph of *xol-1* mRNA levels in WT, *prg-1*, *21ur-4863* deletion, and *21ux-1* deletion worms measured by RT-qPCR. *actin* mRNA served as the internal control. Data expressed as mean ± s.d. of three experiments.

(G) Western blot (anti-FLAG) of GFP::FLAG::XOL-1 (top) levels in WT, *21ur-4863* deletion, and *21ux-1* deletion worms. Alpha-tubulin (bottom) was probed as a loading control.

(H and I) Bar graphs of percent viable (H) and pseudomale (I) progeny of WT, *prg-1*, and *21ur-4863* deletion worms treated with *sex-1(RNAi)*. n > 500 per experimental group. Data expressed as mean ± 2 s.e.m. of three experiments.

See also Figures S5.



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explore potential functions and regulation of germline ncRNA species.

Molecular Cross-talk between Germline Argonaute Pathways

Previous genetic studies have revealed interactions between the Piwi pathway and two Argonaute pathways that propagate epigenetic memories of gene expression states: the WAGO pathway, which targets silenced genes, and the CSR-1 pathway, which targets expressed genes. Targeting by WAGO and CSR-1 Argonautes is readily apparent since both engage 22G-RNAs templated directly from the target RNA by RdRP. Therefore, the comprehensive identification of PRG-1/piRNA target sites affords an opportunity to explore how piRNA targeting correlates with 22G-RNA levels across annotated WAGO and CSR-1 targeted mRNAs.

A striking and unanticipated pattern of 22G-RNA levels emerged from this analysis. On WAGO-targeted mRNAs, piRNA target sites were correlated with three predominant 22G-RNA peaks, one in the center at t12, and one on each side of the targeted site. Interestingly, the central peak at t12 was completely dependent on PRG-1, while the flanking peaks were much less dependent on PRG-1. The flanking peaks that persist in *prg-1* mutants may reflect piRNA-initiated 22G-RNAs that function in WAGO-mediated trans-generational silencing. Consistent with this idea, analyses of data from published WAGO IP experiments indicate that 22G-RNAs at these somewhat *prg-1*-independent flanking sites associate with Argonautes required for propagating piRNA-induced epigenetic silencing (WAGO-1 and WAGO-9) (Figures S2G–S2J). Interestingly, the strongly *prg-1*-dependent 22G-RNAs generated at t12 associate with WAGO-1 only. Thus, it will be interesting to learn why WAGO-1 but not WAGO-9 binds these t12-associated species and whether their biogenesis depends on PRG-1-dependent mRNA slicing which is predicted to occur between t10 and t11.

Our findings also shed light on the relationship between PRG-1 and CSR-1 targeting. Depletion of CSR-1 resulted in an increase in both unique and total piRNA hybrid reads on mRNAs targeted by CSR-1. These findings are consistent with genetic findings that CSR-1 protects its targets from PRG-1-induced silencing (Seth et al., 2013; Wedeles et al., 2013). Moreover, piRNA target regions on CSR-1 target mRNAs exhibit a pattern of 22G-RNA accumulation that is strikingly different from that observed on WAGO-targeted mRNAs. Instead of a central peak and twin flanking peaks, as in WAGO targets, a small but reproducible 22G-RNA peak, positioned just 5 nucleotides 3′ of the piRNA target site (Figures S2K and S2L), was evident in CSR-1 target mRNAs. The 22G-RNA distribution around piRNA target sites in CSR-1 mRNAs remained unchanged after the short period of CSR-1 depletion. This finding suggests that the effect of CSR-1 depletion on patterns of WAGO 22G-RNA accumulation, if any, is less rapid and perhaps less direct than its effect on piRNA targeting of these mRNAs. Unfortunately, depletion of CSR-1 leads to adult sterility, precluding a longer-term multi-generational analysis of 22G-RNA patterns. Taken together, our findings suggest that CSR-1 protects its targets from piRNA silencing in two ways; first by reducing the frequency of PRG-1 piRISC binding, and second, perhaps more indirectly, by preventing 22G-RNA accumulation at t12 and flanking regions correlated with WAGO-1 and WAGO-9 targeting.

Rules Governing piRNA Targeting

Our analysis of base-pairing interactions between piRNAs and their targets suggests that animal Piwi- and AGO-clade Argonautes have broadly similar patterns of targeting. As previously described for miRNA RISC, we find that piRISC function strongly depends on pairing in the seed region and to a lesser extent on 3′ supplemental pairing (Shin et al., 2010). The most significant difference we observe is a shift in 3′ supplementary pairing from positions 13 to 16 in miRISC to positions 15 to 18 in PRG-1 piRISC (Grimson et al., 2007), perhaps consistent with structural differences between miRISC and piRISC (Matsumoto et al., 2016a).

In addition to base-pairing interactions, both AGO and Piwi Argonautes make direct contact with their target RNAs, including specific amino acid contacts with the t1 nucleotide. Human AGO2 and insect Piwi proteins (i.e., Siwi and Aubergine) exhibit a strong preference for adenosine at t1 (t1A), which differs from our finding that PRG-1 prefers t1C. This preference for C may help ensure that PRG-1 target sites often have optimal positioning of a C residue that can serve as a start site for RdRP-dependent amplification of 22G-RNAs. A comparison of the region in PRG-1 that corresponds to the t1 binding pocket in other Argonautes suggests a possible structural basis for this discrimination for t1C. Whereas the polar hydrophobic amino acid Thr640 in Siwi and Aubergine is thought to bind t1A (Matsumoto et al., 2016a), the corresponding position in PRG-1 is a non-polar hydrophobic leucine (Figure S3E).

Using a sensitive epigenetic silencing assay, we were able to directly validate the importance of pairing at each position of the seed and 3′ supplemental pairing regions. Silencing was most sensitive to the loss of pairing at positions 2 and 3, suggesting that targeting is initiated by the first half of the seed region. Remarkably, with the exception of positions 9 to 13, which had very little effect

Figure 6. CSR-1 Prevents piRNA Binding to Its Targets

(A) Box-and-whisker plots of piRNA binding site density in 3,820 CSR-1 targeted mRNAs identified by CLASH in both WT and CSR-1^{depleted} worms. piRNA binding density expressed as the number of normalized CLASH reads per kilobase (target mRNA length) per million mapped reads (RPKM) in WT or CSR-1^{depleted}. Outliers were removed. Mean, median, and p values are indicated. CLASH reads normalized as described in STAR Methods.

(B) Scatterplot of unique piRNA binding site counts per gene for each CSR-1 target in CSR-1^{depleted} versus WT. For 1466 CSR-1 targets, the number of piRNA binding sites increased > 2-fold after depleting CSR-1. A representative target, *dhc-1*, shown in panel (C).

(C) Distribution of *dhc-1* chimeric reads from WT and CSR-1^{depleted} worms.

(D) Scatterplot of the change in mRNA abundance between WT and CSR-1^{depleted} worms versus the change of piRNA binding density for 3,820 CSR-1 targets.

(E) Boxplots of the change in mRNA expression levels between WT and CSR-1^{depleted} worms for 5 sets of genes with different levels of piRNA binding site changes. Median and mean indicated. Each set is significantly different from the prior one, as indicated by the p values.

See also Figures S6.

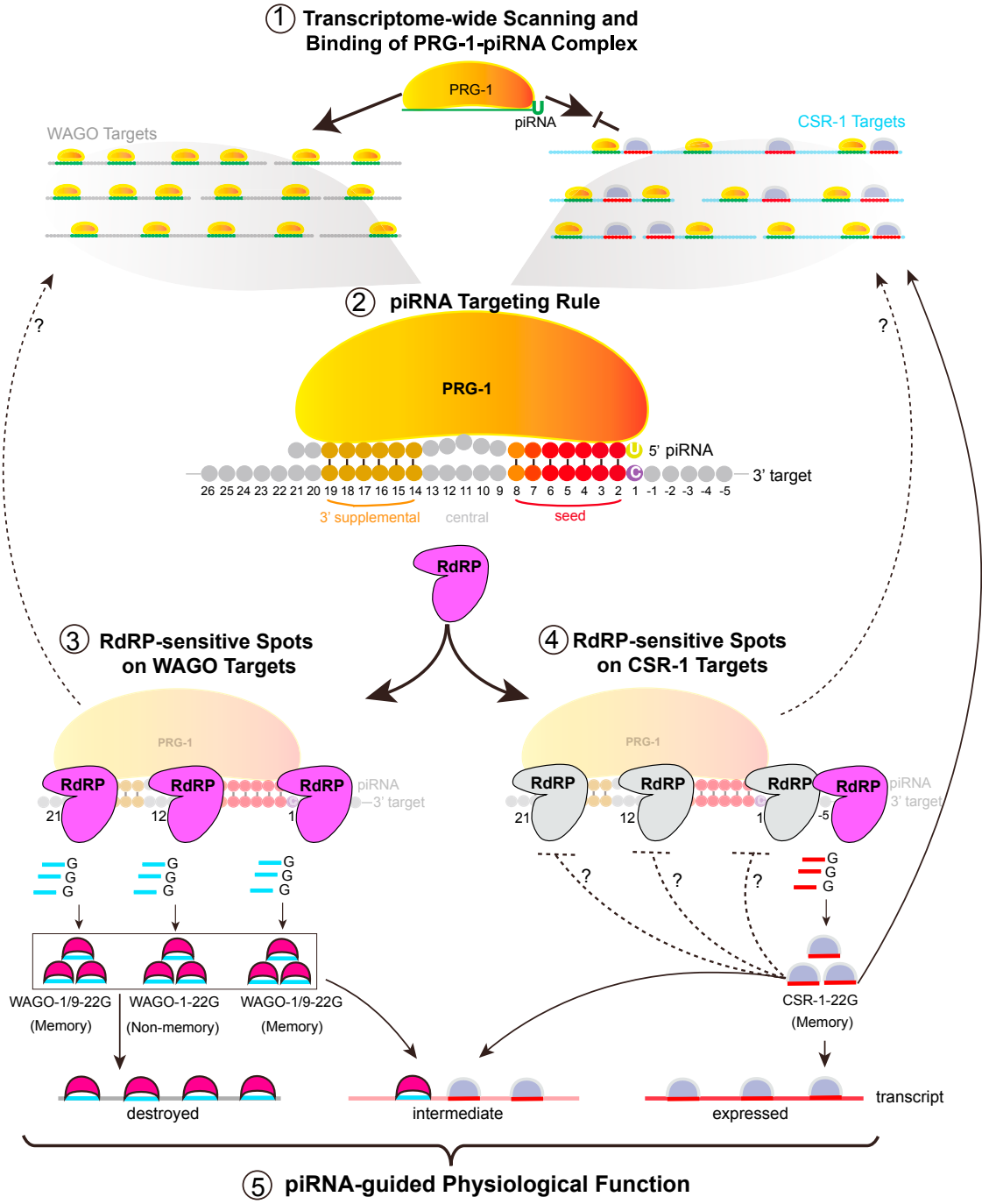


Figure 7. Model for a Regulatory Landscape of piRNAs in the *C. elegans* Germline

on silencing, single nucleotide substitutions at any other location from positions 2 to 8 or 14 to 21 dramatically reduced silencing over the first several generations. Mutants with mismatches in the 3′ supplementary pairing region eventually silenced the target in later generations, but mutants with mismatches in the seed region, especially at g2 and g3, never exhibited full silencing of the target. Thus, seed and 3′ supplementary pairing are of key importance to piRNA targeting. Even single-nucleotide changes dramatically reduced targeting and extended the number of generations required for penetrant silencing.

The Physiology of piRNA Targeting

In most animals, Piwi mutants are completely sterile, likely due at least in part, to loss of transposon regulation. In worms,

most transposons appear to be silenced by epigenetic mechanisms—i.e., WAGO 22G-RNAs and heterochromatin pathways—that maintain transgenerational silencing downstream of PRG-1 (Ashe et al., 2012; Bagijn et al., 2012; Gu et al., 2009; Lee et al., 2012; Shirayama et al., 2012). This additional layer of epigenetic silencing may explain why *prg-1* mutants exhibit relatively minor transposon activation and fertility defects during early generations, but exhibit declining fertility over multiple generations (i.e., a mortal germline phenotype) (Simon et al., 2014).

PRG-1 is nevertheless constantly required to maintain silencing at some loci. Transgenes exposed to both positive (i.e., CSR-1-dependent) and negative (i.e., piRNA-dependent) signals can achieve a balanced state of regulation, where PRG-1 targeting becomes essential to maintain silencing (Seth et al., 2018). At least a few hundred endogenous mRNAs are significantly upregulated in *prg-1* mutants, with a concomitant loss of robust 22G-RNAs levels. One such gene, *xol-1*, is silenced in the hermaphrodite germline by an X chromosome expressed piRNA, *21ux-1* (Seth et al., 2018; Tang et al., 2018). Silencing of *xol-1* ensures that hermaphrodite offspring respond robustly to signals that initiate dosage compensation and sex determination in the early embryo. Although *21ux-1* is by far the most abundant piRNA species, a piRNA with average abundance (*21ur-4863*) binds *xol-1* more efficiently based on the frequency of CLASH hybrid identification. 21ur-4863 is predicted to bind *xol-1* with higher binding energy than predicted for *21ux-1,* highlighting the importance of binding energy rather than abundance in driving piRNA targeting. Surprisingly, both *21ur-4863* and *21ux-1* are required to maintain *xol-1* silencing, suggesting that they—and perhaps other—piRNAs cooperatively silence *xol-1*. Consistent with this idea, the pattern of 22G-RNA induction along *xol-1* extends beyond the regions proximal to these two piRNA target sites, suggesting that additional piRNAs likely contribute to the cooperative regulation of *xol-1* mRNA. Indeed, our CLASH experiments identified 40 piRNAs that target different sites in *xol-1* mRNA. Similarly, because we tagged the endogenous *prg-1* gene with GFP and FLAG to permit tandem-affinity purification, we were able to identify 92 different piRNAs that target sites distributed along the length of *gfp* (Table S1). Cooperative targeting by these piRNAs could explain why 22G-RNA accumulation occurs broadly along silenced *gfp* transgenes (Shirayama et al., 2012; Seth et al., 2018). Remarkably, even though multiple piRNAs regulate *xol-1*, changing a single nucleotide within the seed or 3′ supplementary pairing regions of *21ux-1* can disrupt silencing of *xol-1* and thus affect the regulation of dosage compensation and sex determination.

In summary, our findings show that piRNAs target the entire germline transcriptome. Together with findings from previous and parallel studies our findings also suggest that piRNAs are remarkably versatile in their control of gene expression. piRNAs can act decisively in one generation to initiate epigenetic silencing that persists for multiple generations without need for further piRNA targeting. piRNAs can act cooperatively to silence germline mRNAs (e.g., *xol-1*) that would otherwise reactivate in each generation. And finally, piRNAs can act gradually, over multiple generations, to progressively silence a germline mRNA. Understanding how piRNAs achieve these nuanced modes and tempos of regulation may shed light on whole new vistas of post-transcriptional and epigenetic regulation in animal germlines.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures and three tables and can be found with this article online at https://doi.org/10.1016/j.cell.2018.02.002.

ACKNOWLEDGMENTS

We thank P. Zamore and V. Ambros for suggestions; members of Mello and Ambros labs for discussions, D. Conte for comments and edits on the text; W. Gu for providing the reagents for preparing RNA seq libraries; M. Carmell for proofreading; E. Kittler, D. Wilmot and the Umass Deep Sequencing Core for offering high-throughput sequencing; Caenorhabditis Genetics Center (CGC) for providing strain. This work was supported by an NIH grant (HD078253) to Z. W.; NIH Pathway to Independence Award (GM124460) to W.T.; and NIH grants (GM058800 and HD078253) to C.C.M. C.C.M. is a Howard Hughes Medical Institute Investigator.

AUTHOR CONTRIBUTIONS

Conceptualization, E.S. and C.C.M.; Investigation, E.S., H.C., A.R.O., S.T., M.S., W.T., Y.D., S.D., Z.W. and C.C.M.; Writing-Original draft, E.S. and C.C.M.; Writing-Review & Editing, E.S. and C.C.M.; Supervision, C.C.M and Z.W.

DECLARATION OF INTEREST

The authors declare no competing interests.

Received: January 2, 2018
Revised: January 26, 2018
Accepted: January 31, 2018
Published: February 15, 2018

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rat monoclonal anti-tubulin	BioRad	Cat# MCA77G
Mouse monoclonal anti-GFP	Roche	Cat# 11814460001
Mouse monoclonal anti-FLAG	Sigma-Aldrich	Cat# F3165
Mouse monoclonal anti-mini-AID-tag	MBL	Cat# M214-3
Anti-rat IgG (HRP-conjugated)	Jackson ImmunoResearch	Cat# 712-035-150
Anti-mouse IgG (HRP-conjugated)	Thermo Fisher	Cat# 62-6520 RRID: AB_2533947
Bacterial and Virus Strains		
E. coli: Strain OP50	*Caenorhabditis* Genetics Center	WormBase: OP50
E. coli: Strain HT115	*Caenorhabditis* Genetics Center	WormBase: HT115
E. coli: Ahringer collection	Laboratory of C. Mello	N/A
Chemicals, Peptides, and Recombinant Proteins		
dNTPs	Roche	Cat# 3622614001
Ex Taq DNA Polymerase	TaKaRa	Cat# RR001C
cOmplete EDTA-free Protease Inhibitor Cocktail	Roche	Cat# 11836170001
Proteinase K	New England Biolabs	Cat# P8107S
Ethidium bromide	Sigma-Aldrich	Cat# E1510-10ML
TRI REAGENT	Molecular Research Center, Inc	Cat# TR 118
PIR-1	Gift from Weifeng Gu	N/A
T4 RNA Ligase 1	New England Biolabs	Cat# M0437M
T4 RNA Ligase 2, truncated	New England Biolabs	Cat# M0242L
SuperScript III Reverse Transcriptase	Life Technologies	Cat# 18080085
Turbo DNase	Life Technologies	Cat# AM2238
BsaI	New England Biolabs	Cat# R3535S
Not I	New England Biolabs	Cat# R3189S
TRIS (Base)	avantor	Cat# 4099-06
Sodium dodecyl sulfate (SDS)	Sigma-Aldrich	Cat# L3771-100G
Sodium chloride (NaCl)	Genesee Scientific	Cat# 18-214
Magnesium Acetate Tetrahydrate	Thermo Fisher	Cat# BP215500
Phenol:Chloroform, Low pH	VWR	Cat# 0966.400ml
Isopropyl-β-D-thiogalactoside (IPTG)	Sigma-Aldrich	Cat# 11411446001
Ampicillin	Sigma-Aldrich	Cat# A9518
Tetramisole hydrochloride	Sigma-Aldrich	Cat# L9756-5G
Indole-3-acetic acid (IAA)	Thermo Fisher	Cat# AAA1055614
HEPES	Sigma-Aldrich	Cat# H4034-500G
Sodium Citrate Dihydrate	Thermo Fisher	Cat# BP3271
Triton X-100	Sigma-Aldrich	Cat# T8787-250ML
Tween 20	Fisher	Cat# BP337500
RNase T1	Thermo Fisher	Cat# EN0542
NP-40	EMD Millihpore	Cat# 492018
DL-Dithiothreitol (DTT)	Sigma-Aldrich	Cat# D0632-10G
AcTEV Protease	Thermo Fisher	Cat# 12575023
UltraPure bovine serum albumin (BSA)	Thermo Fisher	Cat# AM2618
SUPERase In RNase Inhibitor	Thermo Fisher	Cat# AM2696

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
EMPIGEN BB detergent	Sigma-Aldrich	Cat# 30326-50ML
T4 Polynucleotide Kinase	New England Biolabs	Cat# M0201L
rATP	Promega Corporation	Cat# E6011
FastAP Thermosensitive Alkaline Phosphatase	Thermo Fisher	Cat# EF0652
Dimethyl Sulfoxide (DMSO)	Thermo Fisher	Cat# BP231-100
Q5 High-Fidelity DNA Polymerase	New England Biolabs	Cat# M0491S
ATP, [γ-32P]	PerkinElmer	Cat# BLU002A250UC
GlycoBlue Coprecipitant	Thermo Fisher	Cat# AM9515
Acid-Phenol:Chloroform, pH 4.5 (with IAA, 125:24:1)	Thermo Fisher	Cat# AM9720
Alt-R S.p. Cas9 Nuclease 3NLS	Integrated DNA Technologies (IDT)	Cat# 1074181
T4 DNA ligase	Enzymatics Inc	Cat# L603-HC-L
Fast SYBR Green Master Mix	Thermo Fisher	Cat# 4385617
Urea	Thermo Fisher	Cat# AC327380010
Critical Commercial Assays		
*mir*Vana miRNA Isolation Kit	Thermo Fisher	Cat# AM1561
RiboMinus Eukaryote Kit for RNA-Seq	Thermo Fisher	Cat# A1083708
Q5 Site-Directed Mutagenesis Kit	New England Biolabs	Cat# E0554S
Zero Blunt TOPO PCR Cloning Kit	Thermo Fisher	Cat# 450245
NextSeq 500/550 High Output v2 kit (150 cycles)	Illumine, Inc.	Cat# FC-404-2002
RNA Clean & Concentrator-5	Zymo Research	Cat# R1015
Anti-FLAG M2 Magnetic Beads	Sigma-Aldrich	Cat# M8823-1ML
Dynabeads Protein G	Thermo Fisher	Cat# 10004D
Pierce Silver Stain Kit	Thermo Fisher	Cat# 24612
Lumi-LightPLUS Western Blotting Substrate	Sigma-Aldrich	Cat# 12015196001
Hyperfilm ECL	Thermo Fisher	Cat# 45001507
RNA Clean & Concentrator Kit	Zymo Research	Cat# R1016
Deposited Data		
Sequencing dataset for CLASH, small RNA-seq and RNA-seq libraries	This study	NCBI: SPR131397
Sequencing dataset for small RNA-seq in *prg-1* mutant	This study	NCBI: SRR513312
Sequencing dataset for PRG-1 IP-small RNA-seq, repeat I	This study	NCBI: SRR538357
Sequencing dataset for PRG-1 IP-small RNA-seq, repeat II	This study	NCBI: SRR2140770
Experimental Models: Organisms/Strains		
List of *C. elegans* strains	This study	Table S2
Oligonucleotides		
List of Oligonucleotides	This study	Table S3
Software and Algorithms		
Bowtie version 0.12.9	Langmead, Trapnell, Pop, & Salzberg, 2009	http://bowtie-bio.sourceforge.net/index.shtml
BEDtools version 2.25.0	Quinlan & Hall, 2010	http://bedtools.readthedocs.io
BLAT version 35x1	Kent, 2002	https://users.soe.ucsc.edu/~kent/src/
Vienna package version 2.3.5	Lorenz et al., 2011	https://www.tbi.univie.ac.at/RNA
BEAR encoder version 2.0	Mattei, Ausiello, Ferrè, & Helmer-Citterich, 2014	https://github.com/emattei/BEAR_encoder
Affinity propagation	Frey and Dueck, 2007	http://www.psi.toronto.edu/index.php?q=affinity%20propagation
deepTools version 2.5.1	Ramírez et al., 2016	https://deeptools.readthedocs.io
Trimmomatic version 0.33	Bolger et al., 2014	http://www.usadellab.org/cms/index.php?page=trimmomatic

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
STAR version 2.4.2a	Dobin et al., 2013	https://github.com/alexdobin/STAR/releases
Samtools version 0.1.19	Li et al., 2009	http://samtools.sourceforge.net
Htseq version 0.7.2	Anders et al., 2015	https://htseq.readthedocs.io
ggplot2 version 2.1.0	Wickham, 2009	https://CRAN.R-project.org/package=ggplot2
uShuffle	Jiang, Anderson, Gillespie, & Mayne, 2008	http://digital.cs.usu.edu/~mjiang/ushuffle/
DESeq version1.22.1	Anders and Huber, 2010	https://bioconductor.org/packages/release/bioc/html/DESeq.html
GraphPad Prism version 5.00	GraphPad Software	https://www.graphpad.com

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for reagents may be directed to, and will be fulfilled by, the Lead Contact, Craig C. Mello (Craig.Mello@umassmed.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

C. elegans Strains and Bacterial Strains

Strains in this study were derived from the Bristol N2 background, and unless otherwise stated, cultured at 20°C on NGM agar plates seeded with OP50 *E. coli* (Brenner, 1974).

METHOD DETAILS

CRISPR/Cas9 genome editing

Cas9/sgRNA: The online CRISPR DESIGN (http://crispr.mit.edu) tool was used to design single-guide (sg) RNAs. For sgRNA construction in co-CRISPR strategy (Kim et al., 2014), BsaI-cut pRB1017 was used as the recipient sgRNA vector, and annealed sgRNA oligonucleotides with additional nucleotides at the 5′ or 3′ ends were ligated into the vector (Arribere et al., 2014; see sgRNA sequences used in this study in Table S3). For sgRNAs used in the preassembled CRISPR/Cas9 ribonucleoprotein complexes strategy (Paix et al., 2015), we ordered crRNA from Integrated DNA Technologies (IDT) using sgRNA sequence.

Donor: The co-CRISPR strategy was used to tag *prg-1* with *gfp::tev::flag* sequence. We amplified 500 bp upstream and 500 bp downstream of *prg-1* by PCR using N2 genomic DNA as a template and the primers 5′-CGGTAATATCCTGTCTGATCTTGCG-3′ and 5′-GATAGAATAATCGAGCCTGGTGTAA-3′. PCR fragments were cloned into pCR-Blunt II-TOPO vector by ligation. A NotI restriction site was introduced by site-directed mutagenesis into donor constructs immediately after the *prg-1* start code. The *gfp::tev::flag* fragments with NotI restriction overhangs were ligated into NotI-linearized donor constructs using T4 DNA ligase. Donors used for the injection of CRISPR/Cas9 ribonucleoprotein complexes were ordered from IDT, their sequences constitute of sgRNA target and at least 40 bp upstream and 40 bp downstream of sgRNA target.

We used *unc-22* a co-CRISPR marker, in which F1 animals with a twitching phenotype (*unc-22* mutant) are enriched for targeted CRISPR/Cas9-mediated genome editing events (Kim et al., 2014). We used *rol-6* as a co-injection marker when injecting CRISPR/Cas9 ribonucleoprotein complexes (Paix et al., 2015). Genome-edited animals were identified among F1 rollers. All strains are listed in Table S2. The oligo sequences used are listed in Table S3.

Small RNA Library Preparation and analysis

Total RNA was extracted from synchronous adult worms using TRI Reagent (Molecular Research Center), according to the manufacturer's instructions. Small RNAs were extracted from total RNA using mirVana miRNA Isolation Kit (Thermo Fisher Scientific). Samples were pretreated with homemade, recombinant 5′ polyphosphatase PIR-1 to remove the γ and β phosphates from triphosphorylated 5′ ends of 22G-RNAs (Gu et al., 2009). The resulting 5′-monophosphorylated small RNAs were ligated to a 3′ adaptor (5′ rAppAGATCGGAAGAGCACACGTCTGAACTCCAGTCA/3ddC/3′; IDT) using T4 RNA ligase 2. The ligation products were then ligated to 5′ adaptor (rArCrArCrUrCrUrUrUrCrCrCrUrArCrArCrGrArCrGrCrUrCrUrUrCrCrGrArUrCrU) was using T4 RNA ligase 1. Ligation products with 5′ and 3′ linkers were reverse transcribed using SuperScript II. The cDNAs were amplified by PCR, and the libraries were sequenced on an Illumina HiSeq platform at the UMass Medical School Deep Sequencing Core Facility.

Small RNA sequencing data were analyzed as described (Lee et al., 2012). Briefly, both 5′ and 3′ adaptor sequences were trimmed using a custom Perl script. Sequencing reads were sorted into different bins according to barcode sequences, and 21-23nts reads

were mapped to the *C. elegans* genome (Wormbase release WS230) allowing no more than 2 mismatches. To account for variation in sequencing depth between samples, each read was normalized to the total number of reads. Normalized counts were visually observed in the UCSC genome browser. All scripts are available upon request.

mRNA Library Preparation

Total RNA from worms was extracted using TRI Reagent (MRC) and ethanol precipitation. RNA samples were processed as described (Zhang et al., 2012). Briefly, ribosomal RNAs were depleted from 4 μg total RNA using the RiboMinus Eukaryote Kit (Life Technologies, A1083708), and the rRNA-depleted samples were treated with Turbo DNase (Ambion) for 30 min at 37°C. RNAs > 200 nt were enriched using RNA Clean & Concentrator-5 (Zymo Research, R1015), fragmented, and reverse transcribed into cDNA using oligo-dT and SuperScript II. The first-strand cDNA was ligated to 5′ and 3′ adapters (see Table S3) using T4 DNA ligase (600 U/μL, Enzymatics Inc, L603-HC-L) for 30 min at 25°C. The library was amplified with a barcoded PCR primer. Deep sequencing was performed on an Illumina HiSeq 2000.

Immunoprecipitation and RNA Isolation

200,000 synchronous adult worms were homogenized in a FastPrep-24 benchtop homogenizer (MP Biomedicals). Immunoprecipitation was performed as described (Tang et al., 2016). In brief, worm extracts were cleared by centrifugation for 10 min at 14,000 × g. Lysates (20 mg total protein) were incubated with anti-FLAG M2 magnetic beads for 1.5 h at 4°C on a rotator. Beads were washed 3 times with IP buffer containing protease inhibitors for 10 min each wash, and then washed once with wash buffer containing 50 mM Tris·Cl (pH 7.5), 2 mM magnesium acetate, and 150 mM NaCl. Immune complexes were treated with 10 mg/ml proteinase K in 2.5% (w/v) SDS, 200 mM EDTA, 100 mM Tris-HCl (pH 7.5) for 10 min at 50°C. RNAs were extracted with acidic phenol/chloroform (low PH) and precipitated with ethanol. Isolated RNA was treated with PIR-1 and subjected to small RNA cloning.

RT-qPCR

Total RNA was treated with Turbo DNase (Ambion), extracted with TRI Reagent (MRC), and precipitated with ethanol. 500 ng of RNA was used as a template for first-strand cDNA synthesis using random hexamers and SuperScript II Reverse Transcriptase (Invitrogen) according to the manufacturer's instructions. cDNA synthesis was performed in triplicate. Real-time PCR was conducted using Power SYBR Green PCR Master Mix (Applied Biosystems) on the Bio-Rad CFX96 Real-Time PCR Detection System. Primer sets I (CAGCTGGAAATTACCGAGGA and GTTGGCCATGGAACAGGTAG), II (AGGTGATGCAACATACGGAA and CGAGAAGCATTGAAC ACCAT), III (CATGGCCAACACTTGTCACT and GCACGTGTCTTGTAGTTCCC), IV (TCAAAGATGACGGGAACTAC and GCTTCCAT CTTCAATGTTGT) were used to amplify *gfp* transcript. Primers AGCTTCTTCGAGATGCGTTC and CTTGTCGCACACGGTTCTTG were used to amplify *xol-1* transcript. Primers TCGGAGTTCCGATCATCTCG and CAGGGTGACAGCTCTATCGT were used to amplify *comt-3* transcript. Primers TCTTTCTCCGTCTCCTCTCC and GCCATCAGGTGATCCAGAAC were used to amplify *fbxb-97* transcript. Primers GGCCCAATCCAAGAGAGGTATCC and GGGCAACACGAAGCTCATTGTA were used to detect *act-3* mRNA. Error bars in the graph indicate the standard deviation (SD) in all statistical analysis.

Western Blot Analysis

Protein lysates were prepared from synchronous populations of young adult or gravid adult worms. Proteins (50 μg) were separated on precast denaturing polyacrylamide gels (Thermo Fisher Scientific) and transferred to PVDF membranes (Bio-Rad) using a Trans-Blot Turbo Transfer System (Bio-Rad). Membranes were blocked and probed with primary antibodies: monoclonal anti-FLAG (Sigma, M8823), monoclonal anti-GFP (Roche, 11814460001), monoclonal anti-TUBULIN ALPHA (AbD Serotec, MCA77G), and anti-mini-AID-tag (MBL International, M214-3). Primary antibodies were detected using HRP-linked secondary antibodies: donkey anti-rat IgG(Jackson ImmunoResearch, 712-035-150) or goat anti-mouse (Thermo Fisher Scientific, 62-6520).

RNAi

RNAi was performed by feeding worms *E. coli* strain HT115(DE3) transformed with the control vector L4440 or a gene-targeting construct from the *C. elegans* RNAi Collection (Kamath and Ahringer, 2003). Bacteria were grown overnight in LB with antibiotics at 37°C. NGM plates containing 1 mM isopropyl β-d-thiogalactoside and 100 μg/ml ampicillin were seeded with the overnight culture (100 μl per plate) and incubated for 24 hours at 25°C. L1 larvae were placed on RNAi plates at room temperature and the phenotypes of the adults and their F1 progeny were scored.

Microscopy

Transgenic worms expressing mRFP or GFP were anaesthetized in 0.1 mM levamisole (Sigma, 16595-80-5) on glass slides with 10-mm superfrost circles (Thermo Scientific, 3032) and imaged immediately. Epi-fluorescence and differential interference contrast microscopy were performed in a Zeiss Axioplan2 microscope. Images were captured and processed using Zeiss Axiovision software.

Auxin-inducible Depletion of CSR-1

Auxin treatment was performed as described (Zhang et al., 2015a). 100 mM indole-3-acetic acid (IAA; Alfa Aesar, 10171307) was prepared in ethanol and stored for up to one month at 4°C. NGM plates containing 500 μM IAA (prepared by adding IAA to NGM

agar at ~50°C) were seeded with fresh concentrated OP50 and incubated at RT for 48 hours. IAA plates stored at 4°C were warmed to room temperature for 1 hour before use. *aid::csr-1* (WM570) or wild-type worms were placed on IAA plates as L1 larvae and grown to the young adult stage at 20°C.

Viability and Pseudomale Development

Ten gravid hermaphrodites (WT or mutant) were picked onto individual plates, and transferred to new plates every 12 hours. The number of embryos and hatched larvae were counted at each transfer, and the plates were summed to determine the total number of progeny per hermaphrodite. Progeny were allowed to develop at room temperature for several days, and the number and phenotypes of F1 adult animals were scored at room temperature. Viability is expressed as the percentage of F1 progeny that develop into adults. Pseudomales were identified as F1 adult hermaphrodites that develop male-like tails. *sex-1*(RNAi) was used to enhance the sex-determination defect (masculinization) caused by *xol-1* gain of function.

PRG-1 CLASH Protocol

Worm growth, UV-crosslinking, and worm lysis: L1 worms were grown on large NGM plates with concentrated OP50. Synchronous adult worms (~1,000,000) were collected and transferred to 50 large NGM plates without OP50. Plates were placed on ice in a Stratalinker 1800 (Stratagene) and worms were irradiated with 254-nm light at 1.0 J/cm^2 to crosslink protein and RNA.

Irradiated worms were homogenized in cooled lysis buffer composed of 20 mM HEPES-KOH (pH 7.5), 125 mM citrate sodium, 2 mM $MgCl_2$, 1 mM DTT, 0.5% (v/v) Triton X-100, 0.1% (v/v) Tween 20, 0.04% protease inhibitors (Complete Mini EDTA-free; Roche) using the FastPrep-24 homogenizer. Lysates were centrifuged at 14,000 r*cf.* (x g) for 10 min at 4°C, and supernatants were treated with RNase T1 (3 U/μL) for 15 min at room temperature on a rotator.

GFP pull-down and TEV cleavage: Worm lysates were incubated with GBP beads for 2 hr at 4°C, and then beads were washed three times with washing buffer I (50 mM HEPES-KOH, pH 7.5, 300 mM KCl, 0.05% NP-40, 0.5 mM DTT, 0.04% protease inhibitors) for 5 min each, twice in washing buffer II (50 mM HEPES-KOH, pH 7.5, 500 mM KCl, 0.05% NP-40, 0.5 mM DTT, 0.04% protease inhibitors) for 5 min each, and once in TEV buffer (50 mM Tris-HCl pH 8.0, 0.5 mM EDTA, 1 mM DTT, 2 mM MgOAC) for 5 min. Beads were suspended in 100 μL TEV buffer, and 8 μL AcTEV Protease was added to cleave TEV site between the GFP and FLAG tags. The cleavage reaction was incubated for 2 hr at room temperature, releasing FLAG::PRG-1/RNA complexes into solution.

Purification of FLAG::PRG-1/RNA complexes: To capture FLAG::PRG-1/RNA complexes, anti-FLAG antibodies were incubated with Protein G dynabeads (Life Technologies) for 30 min at room temperature. Beads were washed three times with washing buffer I for 5 min each, twice with lysis buffer containing 0.5% BSA for 10 min each, and twice with lysis buffer containing 0.01% BSA for 10 min each. FLAG::PRG-1/RNA complexes were captured by incubating the TEV-released complexes with washed Dynabeads in TEV buffer containing 2% RNase inhibitor for 1 h at 4°C.

5′ end phosphorylation and intramolecular ligation: FLAG::PRG-1/RNA immune complexes were washed once with 1 × PBS, 2% Empigen (PMPG buffer: Sigma, 30326), once with 5 × PBS, 2% Empigen, and once with 1 × PNK buffer. FLAG::PRG-1/RNA complexes were treated with 0.5 U/μl T4 Polynucleotide Kinase in the PNK buffer (1 mM DTT, 1 mM rATP, 1 U/μL RNase inhibitor, 0.005 U/uL Turbo DNase) for 30 min at room temperature, and then washed once with 1 × PMPG buffer, once with 5 × PMPG buffer, and once with 1 × PNK buffer. FLAG::PRG-1-bound, RNAs molecules were ligated using 2 U/μl T4 RNA ligase in 1x ligase buffer (containing 1 mM rATP, 1 U/uL RNase inhibitor) without DTT for 2 hr at 16°C on a rotator. The ligation mixture was washed once with 1 × PMPG buffer, once with 5 × PMPG buffer, and once with 1 × PNK buffer.

RNA dephosphorylation and ^{32}P-labeled 3′ linker ligation: The FLAG::PRG-1/RNA complexes on Dynabeads were incubated with 0.1 U/μL FastAP Thermosensitive Alkaline Phosphatase (Life Technologies, EF0652) in 1 × FastAP buffer containing 1 U/μl RNase inhibitor for 30 min at room temperature on a rotator. Complexes were then washed once each with 1 × PMPG buffer, 5 × PMPG buffer, 1 × PNK buffer, and 1 × RNA ligase buffer without DTT. Dephosphorylated RNA was incubated with 33 uM 3′ linker (~10% ^{32}P-labeled) in 1 × ligation buffer containing 2.5 U/μL RNA ligase, 1 mM rATP, 2.5% DMSO, 15% PEG 8000, and 1 U/μL Rnase inhibitor for 2 hr at 16°C. Beads were washed once each with 1 × PMPG buffer, 5 × PMPG buffer, and 1 × PNK buffer.

Elution of FLAG::PRG-1/RNA complexes, SDS-PAGE, and transfer to nitrocellulose: FLAG::PRG-1/RNA complexes were eluted in 1 × SDS-PAGE loading buffer for 10 min at 70°C. Duplicate samples (one for western blot analysis one to recover complexes with ligated linkers) were separated on a NuPAGE 4%–12% Bis-Tris protein gel (Life Technologies, NP0321) using NuPAGE SDS-MOPS running buffer (Life Technologies, NP0001) in a cold room. FLAG::PRG-1/RNA complexes were transferred to pre-cut nitrocellulose blotting membrane (Invitrogen, LC2000) in a wet-transfer tank with NuPAGE transfer buffer containing 10% methanol, overnight at 30 V. The membrane was cut into two parts along a marker. One part was analyzed by western blot analysis using anti-FLAG antibody as described above. The other part of the membrane was air-dried and exposed to film (Sigma, Z350370) for 2 hr at –80°C. The developed film was aligned to the membrane and ^{32}P-labeled region of the membrane containing FLAG::PRG-1/RNA complexes were excised.

Proteinase K treatment and RNA extraction: Membrane slices were treated with 50 μl of proteinase K (New England Biolabs, P8107S) in proteinase K buffer (100 mM Tris-HCl, pH 7.4, 50 mM NaCl, 10 mM EDTA) for 20 min at 37°C, and then 42% Urea/Proteinase K buffer was added into sample for an additional 20 min at 37°C. RNA was extracted with phenol/chloroform/isoamyl alcohol, and then precipitated with ethanol/isopropanol and 5 μg GlycoBlue (Ambion, AM9516), overnight at –80°C. RNA pellets were washed with 70% cold ethanol and dissolved in RNase-free water.

5′ phosphorylation, RNA isolation, and 5′ linker ligation: RNAs were incubated with 0.5 U/μL T4 Polynucleotide Kinase in 1 × PNK buffer (1 mM ATP, 1 U/μL RNase inhibitor) for 30 min at 37°C. RNA was purified using an RNA Clean & Concentrator Kit (Zymo Research, R1016) according to the manufacturer's instructions. RNA was ligated to 10 uM 5′ linker using 2 U/μL RNA ligase in 1 × ligation buffer containing 1 mM ATP, 2.5% DMSO, 8% PEG 8000, 1 U/μL RNase inhibitor for 3 hr at 16°C.

RNA extraction, reverse transcription, and library construction: 5′-ligation products were purified using RNA Clean & Concentrator Kit, and converted to first-strand cDNA using primer (GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT) with Superscript III Reverse Transcriptase (Life Technologies, 18080-093), according to the manufacturer's instructions. cDNA was amplified using linker primers and Q5 High-Fidelity DNA Polymerase (New England Biolabs, M0491S), according to the manufacturer's instructions. PCR products were separated on a 12% polyacrylamide gel in 1 × TBE buffer and stained with ethidium bromide. Gel slices containing amplified libraries (160 to 190 bp) were excised and the library was extracted using a Gel Extraction Kit with columns (QIAGEN).

Bioinformatic Analysis of CLASH Data

Pipeline for mapping CLASH hybrids: CLASH chimeras containing full-length piRNA sequences were initially selected from the library using Bowtie, version 0.12.9 (Langmead et al., 2009). piRNAs were then trimmed from the CLASH chimeras, and the remaining sequences were mapped to the *C. elegans* genome (Wormbase WS230) (Yook et al., 2012). These candidate piRNA-target sequences were then classified based on annotations using a customized Perl script and BEDtools (version 2.25.0) (Quinlan and Hall, 2010). As annotation is missing in this release, lincRNAs and Tc1 and Tc3 transposons were manually annotated by mapping their sequence to the genome using BLAT (version 35x1) (Kent, 2002).

piRNA-target duplex prediction, CLASH chimera refinement, and clustering: piRNA-target duplexes and changes in Gibbs free energy (ΔG) were calculated using RNAfold in Vienna RNA Package (version 2.3.5) (Lorenz et al., 2011). Based on the initial alignment, additional bases were added to the 5′ or 3′ ends of trimmed target sequences, based on longer CLASH chimeras or on the reference genome. The duplex prediction and ΔG were recalculated for these refined target sites.

Matching relationships for piRNA-mRNA chimeras detected at least 5 times were summarized as follows Alignments from predicted duplexes were converted into matches, mismatches, and internal bulges using BEAR encoder (Mattei et al., 2014). The piRNA side of the duplex was clustered using affinity propagation (Frey and Dueck, 2007) with a preference score of 5000. The similarity score for clustering was defined as a negative edit distance between duplexes, ranging from 0 to –21. The cluster plot was then generated using deepTools (version 2.5.1) (Ramírez et al., 2016).

Conservation: Pre-calculated phyloP scores for the *C. elegans* genome (version WS220) based on 7 Caenorhabditis species were downloaded from UCSC browser (Kent et al., 2002; Pollard et al., 2010). 9400 coding genes fully covered with phyloP score were used. The piRNA target sites on these genes were separated into 3 groups according to their starting codon position. Then, the average phyloP scores for genomic positions that are on, around, or off the target site were calculated.

RNA-seq data processing and differentially expressed gene analysis: Adaptor sequences and low-quality reads were removed using Trimmomatic, version 0.33 (Bolger et al., 2014). The short reads that passed quality control were mapped to the genome using STAR, version 2.4.2a (Dobin et al., 2013). The read counts for each gene with uniquely mapped strand-specific reads were calculated using Samtools, version 0.1.19 (Li et al., 2009) and HTseq, version 0.7.2 (Anders et al., 2015). Differential expression analyses were performed using DESeq version 1.22.1 (Anders and Huber, 2010).

22G-RNAs around the piRNA target sites: Adaptor sequences and low-quality reads were removed using Trimmomatic (version 0.33). The short reads were then mapped to the genome using STAR (version 2.4.2a) and normalized to sequencing depth. Short reads 21- to 23-bp long with 5′ G were defined as 22G-RNAs. Their 5′ end signals were aggregated for each position around the target sites and normalized to the number of transcriptomic C at these positions.

Normalization of CLASH counts: The number of unique CLASH chimeras in each library was normalized to the total number of unique tRNA-piRNA chimeras, rRNA-piRNA chimeras, tRNA fragments, or rRNA fragments across four replicate datasets. Structural RNAs were used because they are more likely to represent degradation products, and less likely to be influenced by piRNAs. Normalization was also performed using the total number of piRNA chimeras. Each normalization method produced similar results. The data normalized to tRNA-piRNA chimeras is presented.

Scatterplots: Scatterplots were generated using ggplot2 (Wickham, 2009).

CSR-1 and WAGO targets: The 4176 CSR-1 target genes and 1118 WAGO target genes were previously defined (Claycomb et al., 2009; Gu et al., 2009).

Soma-specific genes: modENCODE RNA-Seq data has defined 2423 soma-specific genes expressed in young adult worms (Li et al., 2014).

piRNA target sequence shuffling: piRNA target sequences were shuffled while maintaining their di-nucleotide frequencies using uShuffle (Jiang et al., 2008).

QUANTIFICATION AND STATISTICAL ANALYSES

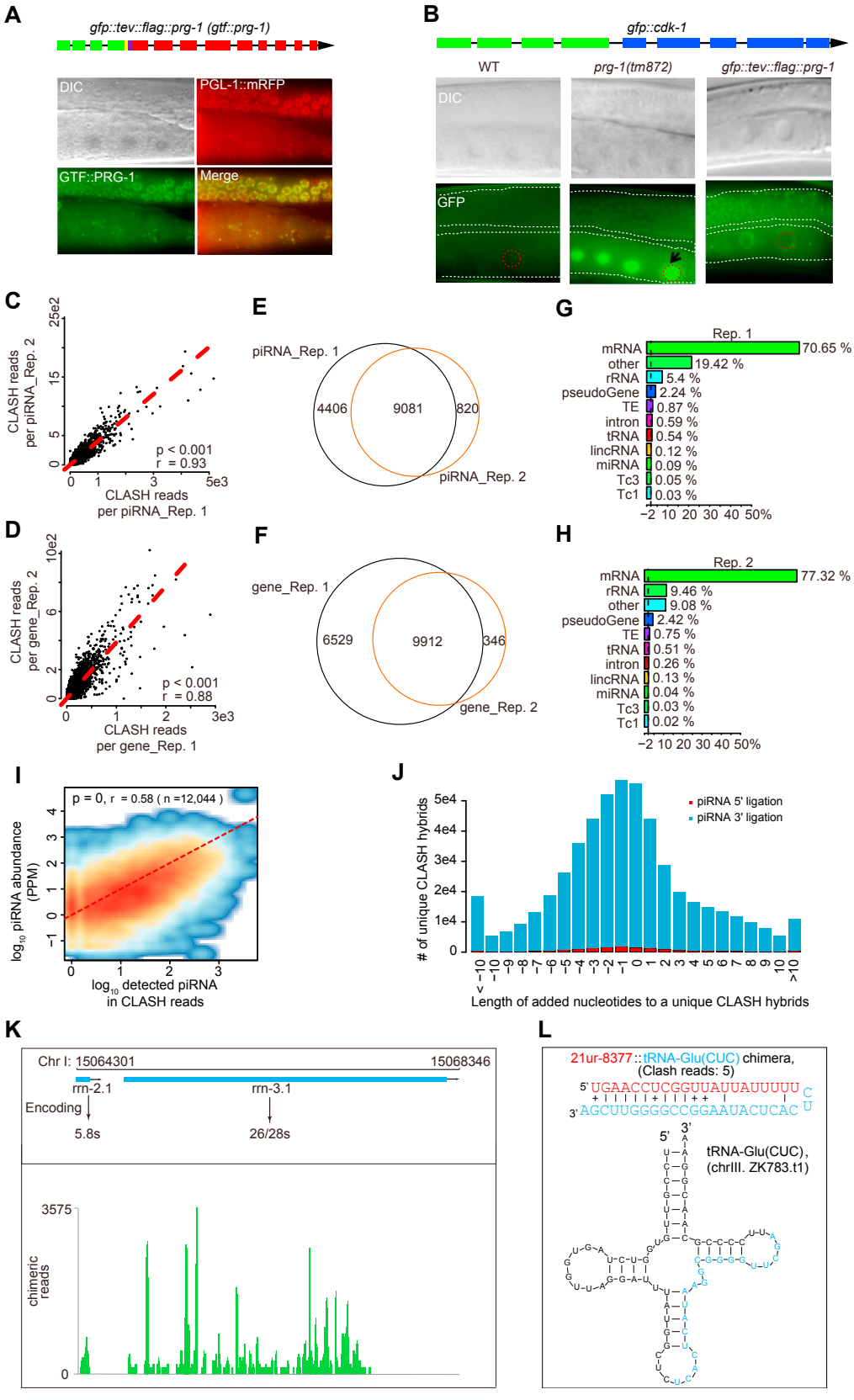
piRNAs abundance was defined as the average normalized rank calculated from two independent PRG-1 IP libraries. We then classified unique CLASH target sites into three groups (≤33%, 33%–66%, and > 66%) based on their piRNA abundance (Figure 1I). The

CSR-1 targets were classified into five groups by log2 change in piRNA binding density between CSR-1 depleted and wild-type (Figure 6E). The p values between adjacent groups was estimated using a two-sided t test. The standard error of the mean (s.e.m) in Figures 4B and 4C; 5C, D, H, and I; S4C, G, and H was calculated using using GraphPad Prism (version 5.00 for Windows; GraphPad Software, La Jolla California USA, https://www.graphpad.com). All other statistical analyses in this study were performed in R (version 3.2.2). The p values in Figures 1G, 1I, 6A, and 6E were calculated using function t.test() with all default parameters. The s.e.m. in Figure 3E was estimated using function sd() and sqrt(). The Pearson correlation coefficient in Figure 6D was estimated using function cor() with all default parameters. Linear regressions between replicates (Figures S1C and S1D) and between piRNA abundance in wild-type and in CLASH libraries (Figure S1I) were performed using function lm() with all default parameters. The corresponding p values and Pearson correlation coefficients were extracted from the linear models.

DATA AND SOFTWARE AVAILABILITY

The two PRG-1 IP libraries used to rank piRNA abundance can be found under NCBI: SRR538357 and SRR2140770. Small RNA-seq library from *prg-1* mutant background is available from NCBI: SRR513312. All other data used in this study is available from NCBI: SRP131397. All scripts for this study are available upon request.

Supplemental Figures



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Figure S1. Two Independent CLASH Datasets, Related to Figure 1

(A) Schematic of the CRISPR-engineered *gfp::tev::flag::prg-1* (*gtf::prg-1*) gene (top), and fluorescence micrographs (bottom) of GFP::TEV::FLAG::PRG-1 (green) expression in the germline. PGL-1::mRFP (red) is a constitutive P granule marker.

(B) Schematic of the *gfp::cdk-1* transgene (top), and fluorescence micrographs (bottom) of GFP::CDK-1 (green) expression in oocyte nuclei of wild-type (WT), *prg-1(tm872)*, and *gtf::prg-1* worms. *prg-1(tm872)* is a loss-of-function allele that activates GFP::CDK-1 expression in (arrowheads). The GTF::PRG-1 fusion is functional.

(C and D) The graphs present the number of reads mapped to each identified piRNA (C) and mRNA (D) in one experiment relative to the other one. r, Pearson's correlation coefficient.

(E and F) Venn diagram summarizing the overlap in the number of piRNAs (E) and mRNA (F) in two replicates.

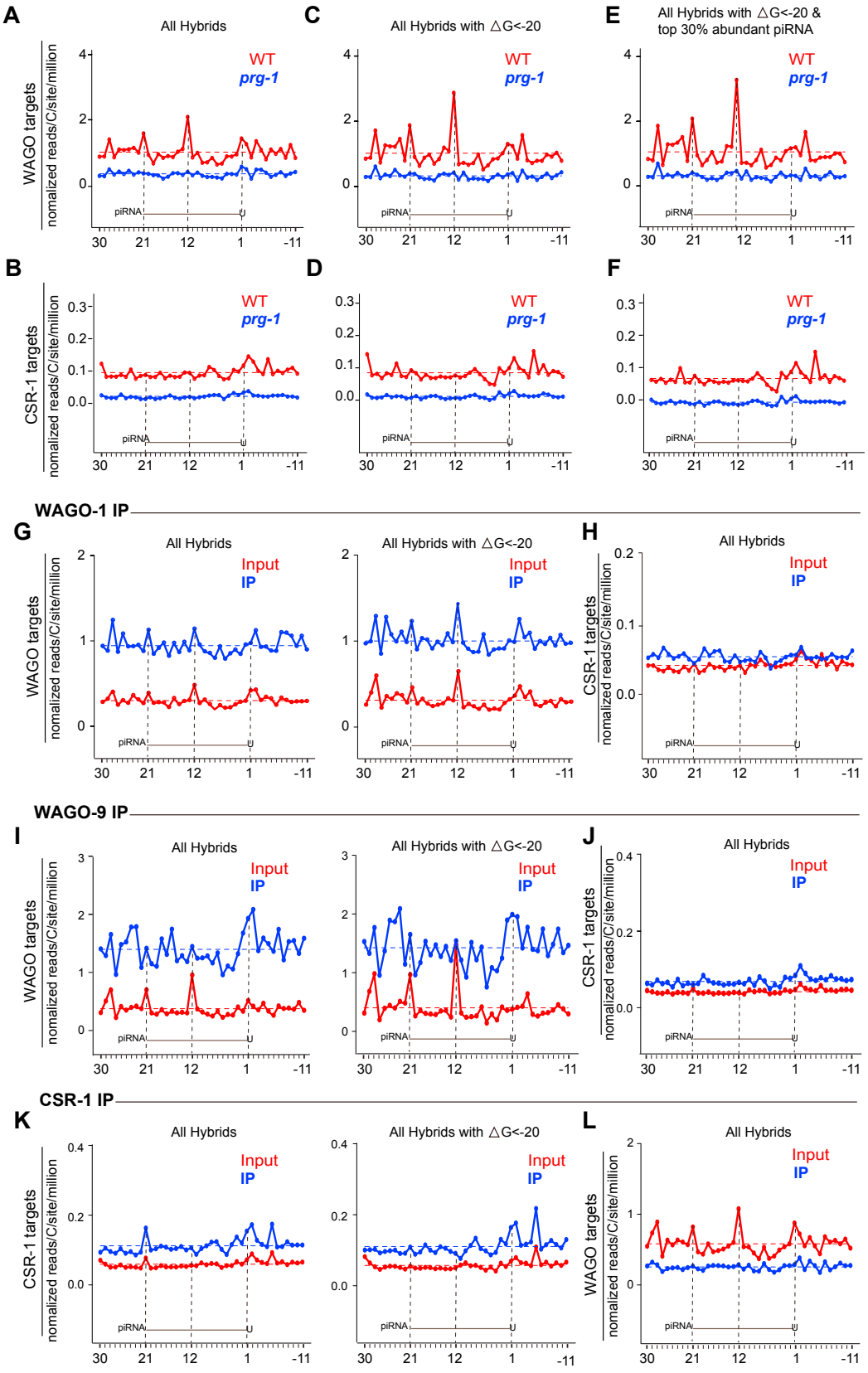
(G and H) Distribution of all piRNA interactions among various types of RNAs in two replicates. mRNAs are the main piRNA targets and represent more than 70%.

(I) Correlation between the abundance of CLASH-recovered piRNAs (x axis) and overall piRNA abundance (y axis).

(J) Histogram showing the number of nucleotides added to the 5′ (negative) or 3′ (positive) ends of CLASH-recovered target regions to create "ideal" piRNA/target RNA pairs. "0" indicates that no nucleotide was added to the target sequence. Hybrids ligated to the 3′ ends of piRNAs (blue), and hybrids ligated to the 5′ ends of piRNAs (red).

(K) Distribution of chimeric *rrn-2.1* and *rrn-3.1* reads (green) identified by CLASH and genomic locus on the top (blue).

(L) Putative interactions between piRNA and tRNA. *21ur-8377* reproducibly bind to the same region of tRNA-Glu(CUC), marked blue on the tRNA structure (chrIII. ZK783.t1).



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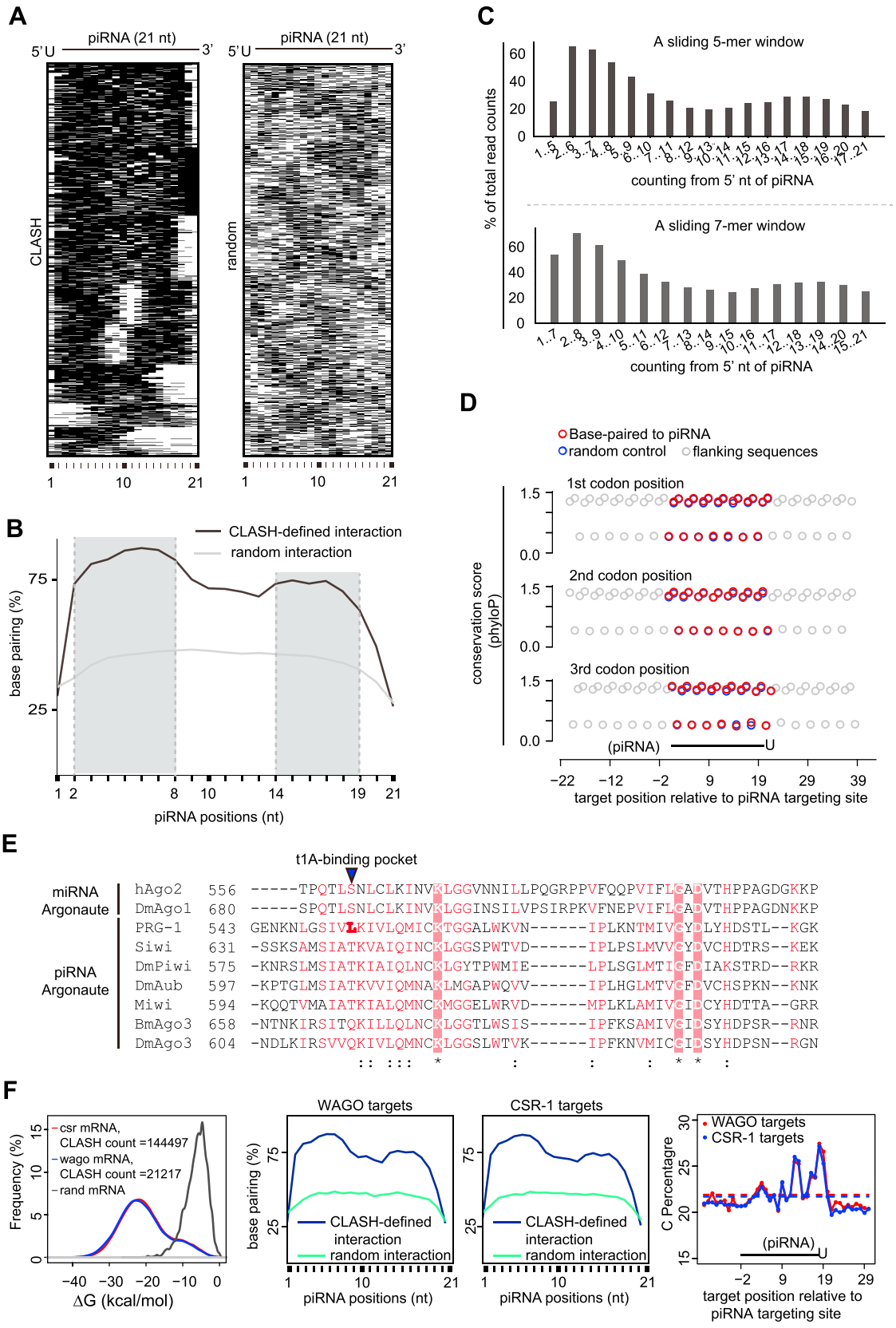
Figure S2. 22G-RNAs Signal in the piRNA Binding Region, Related to Figure 2

(A–F) 22G-RNAs peak at the center and ends of piRNA binding sites in another dataset, similar to Figure 2.

(G and H) Distribution of 22G-RNA signal in input versus WAGO-1 IP for WAGO targets (G), and for CSR-1 targets (H).

(I and J) Distribution of 22G-RNA signal in input versus WAGO-9 IP for WAGO targets (I), and for CSR-1 targets (J).

(K and L) Distribution of 22G-RNA signal in input versus CSR-1 IP for CSR-1 targets (K), and for WAGO targets (L).



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Figure S3. Features of piRNA and Target Interactions, Related to Figure 3

(A) Positions of base-paired nucleotides in piRNAs for all piRNA:mRNA interactions. piRNA:targets duplex structure predictions were calculated using RNAfold, and clustered using APcluster. Shuffled interactions (targets swapped between piRNAs) served as random control. Base-paired nucleotides are shown along the piRNA length.

(B) Hybridization profile showing all interactions. The predicted base-pair frequency at each piRNA position is plotted along the piRNA length.

(C) A 5-mer and 7-mer sliding window search showing preferential seed and 3′ supplementary pairing between piRNAs and target sites. Similar analysis to the analysis in Figure 2C.

(D) Average mRNA conservation level on the piRNA-miRNA contact sites. *C. elegans* phyloP score starting from each codon position at the target sites. See also STAR Methods.

(E) Multiple sequence alignment of the t1A binding pocket region of Argonaute proteins (Matsumoto et al., 2016a). The figure was prepared using ESPript3 (http://espript.ibcp.fr/ESPript/ESPript).

(F) Comparison of base paring energy, base pairing pattern and Cytosine percentage for WAGO and CSR-1 targets. Left: the predicted binding energies between piRNAs and target sites for WAGO and CSR-1 targets and between randomly matched pairs are distributed, respectively. Middle: the predicted frequency of a piRNA position to be base paired is separately plotted for WAGO and CSR-1 target. Right: the percentage of C base around piRNA targeting site in WAGO and CSR-1 targets.

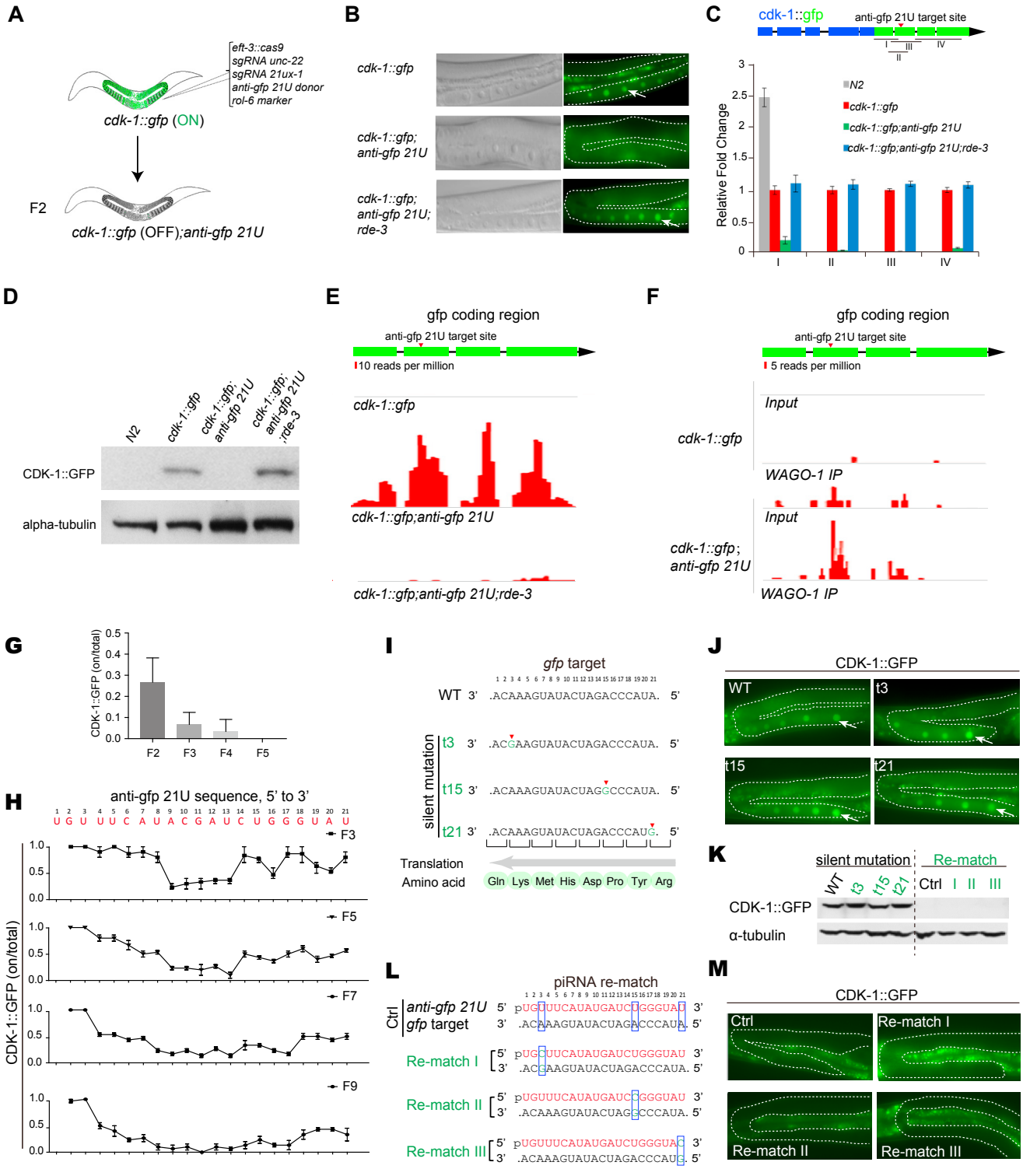


Figure S4. *anti-gfp piRNA* Silences Its Target *cdk-1::gfp*, Related to Figure 4

(A) Generation of CRISPR-Cas9-mediated *anti-gfp* piRNA knock-in strain.

(B) CDK-1::GFP expression in *cdk-1::gfp*, *cdk-1::gfp; anti-gfp* piRNA and *cdk-1::gfp; anti-gfp* piRNA*; rde-3* worms. *rde-3* encodes a nucleotidyltransferase required for the production of piRNA-initiated 22G-RNAs. CDK-1::GFP fluorescence signals in the germline nuclei denoted by arrowheads. Dashed lines outline the position of germline. Bright signals outside of the germline are from intestinal autofluorescence.

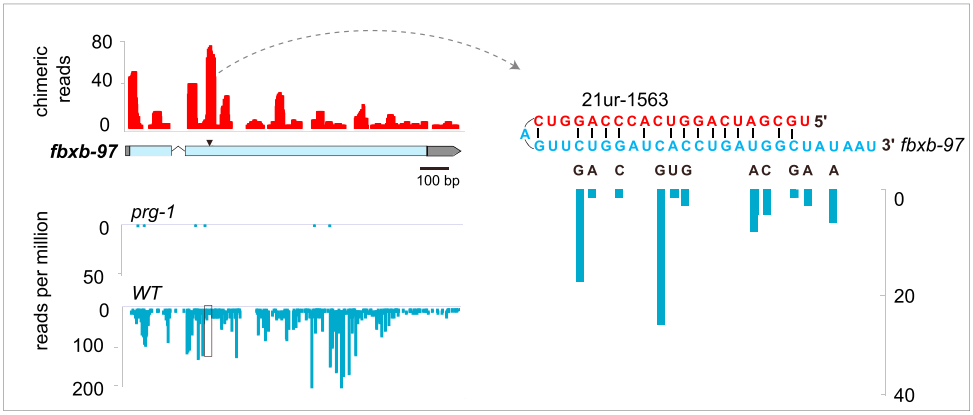
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(C) RT-qPCR analysis of *cdk-1::gfp* mRNA levels in N2, *cdk-1::gfp*, *cdk-1::gfp; anti-gfp* piRNA, and *cdk-1::gfp; anti-gfp* piRNA*; rde-3* worms. Schematic of the single-copy transgene *cdk-1::gfp*. Red arrowhead indicates the *anti-gfp* piRNA target site. The black line beneath the schematic represent the location of PCR product. Data expressed as mean ± s.d. of three experiments.
(D) western blot analysis of CDK-1::GFP in N2, *cdk-1::gfp*, *cdk-1::gfp*; *anti-gfp* piRNA, and *cdk-1::gfp; anti-gfp* piRNA*; rde-3* worms. Tubulin is used as loading control.
(E) Schematic showing level of 22G-RNAs targeting *gfp* in *cdk-1::gfp*, *cdk-1::gfp; anti-gfp* piRNA and *cdk-1::gfp; anti-gfp* piRNA*; rde-3* worms.
(F) *anti-gfp* piRNA-induced 22G-RNAs were loaded into WAGOs.
(G and H) Similar analysis to Figures 4B and 4C, graphs showing the fraction of GFP-positive worms in the presence of perfectly complementary *anti-gfp* piRNA with full match at each position at the F2, F3, F4 and F5 generations (G), or *anti-gfp* piRNA with single-nucleotide mismatches at each position at the F3, F5, F7 and F9 generations (H). Data expressed as mean ± 2 s.e.m. of three experiments.
(I) Diagram showing silent mutations in *gfp* that create single-nucleotide mismatches with *anti-gfp* piRNA at positions t3, t15, and t21 (marked by green).
(J) *cdk-1::gfp* worms with silent mutations in *gfp* maintain WT levels of expression.
(K) western blot showing CDK-1::GFP protein levels in worms with silent mutations in *gfp* (left). *anti-gfp* piRNA mutations that restore base pairing (re-match, right) confer silencing.
(L) Nucleotides (green) showing piRNA re-match.
(M) Fluorescence micrographs showing the silence of CDK-1::GFP expression by piRNA re-match, see also Figure S4K (right).

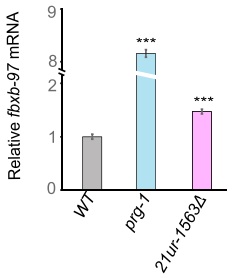
A

	Genotype	*sex-1* RNAi dead eggs (%) (Number of Animal Scored)	*sex-1* RNAi Dumpy animals (%) (Number of Animal Scored)
Rep. 1	WT	43.9 (305)	6.9 (504)
	prg-1(tm872)	8.2 (461)	15.8 (322)
	21ur-4863Δ	13.0 (362)	18.3 (316)
Rep. 2	WT	54.8 (418)	4.6 (241)
	prg-1(tm872)	7.0 (316)	22.9 (297)
	21ur-4863Δ	14.8 (330)	23.2 (185)
Rep. 3	WT	45.6 (226)	9.5 (220)
	prg-1(tm872)	11.0 (218)	25.8 (190)
	21ur-4863Δ	13.7 (190)	15.5 (168)

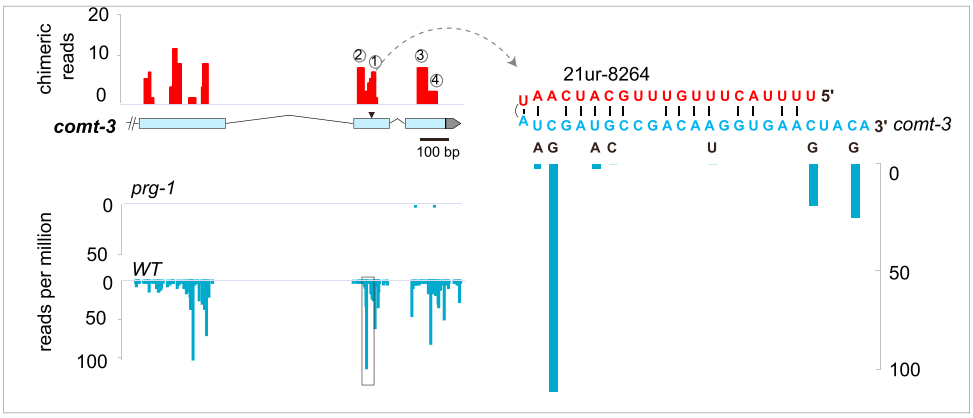
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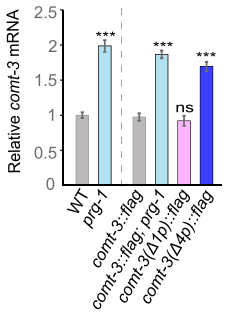
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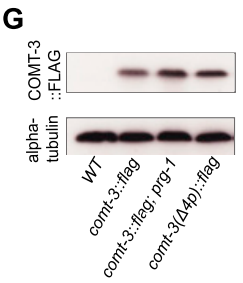
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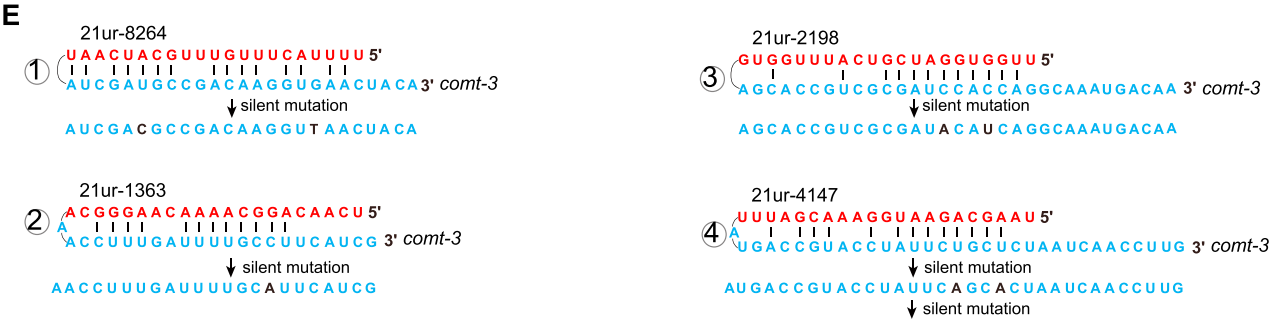
F



G



E



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Figure S5. Analysis of piRNAs Targeting *xol-1*, *fbxb-97* and *comt-3*, Related to Figure 5

(A) Percentage of dead egg and dumpy animals in WT, *prg-1 (tm872)*, and *21ur-4863* deletion strains after *sex-1* RNAi treatment.

(B) Distribution of *fbxb-97* chimeric reads (red) identified by CLASH, and distribution of 22G-RNAs (blue) in *prg-1 (tm872)* and WT. The sequence and base pairing of *21ur-1563::fbxb-97* chimera are shown (right). The piRNA target site location is indicated by the inverted black triangle.

(C) RT-qPCR analysis of *fbxb-97* mRNA level in WT, *prg-1 (tm872)*, and *21ur-1563* deletion worms. actin mRNA served as the internal control. Data were collected from three independent biological replicates. Error bars represent standard deviation.

(D) Distribution of *comt-3* chimeric reads (red) identified by CLASH, and distribution of 22G-RNAs (blue) in *prg-1 (tm872)* and WT. The sequence and base pairing of *21ur-8264::comt-3* chimera is shown (right). The piRNA target site location is indicated by the inverted black triangle.

(E) Diagram of silent mutations in four different piRNA target sites in *comt-3*. Locations of sites indicated in Figure S4D.

(F) Bar graphs showing the change of *comt-3* mRNA level in WT, *prg-1 (tm872)*, *comt-3::flag*, *comt-3::flag;prg-1 (tm872)*, *comt-3(Δ1p)::flag* and *comt-3(Δ4p)::flag* animals. *Δ*1p represents silent mutations in *comt-3* to create mismatches with *21ur-8264*. *Δ*4p represents silent mutations in *comt-3* that create mismatches with *21ur-8264*, *21ur-1363*, *21ur-2198* and *21ur-4147*. As shown in Figure S5E. Data expressed as mean ± s.d. of three experiments.

(G) western blot analysis of COMT-3::FLAG level in WT, *comt-3::flag*, *comt-3::flag;prg-1 (tm872)* and *comt-3(Δ4p)::flag* animals. Tubulin were used as loading control.

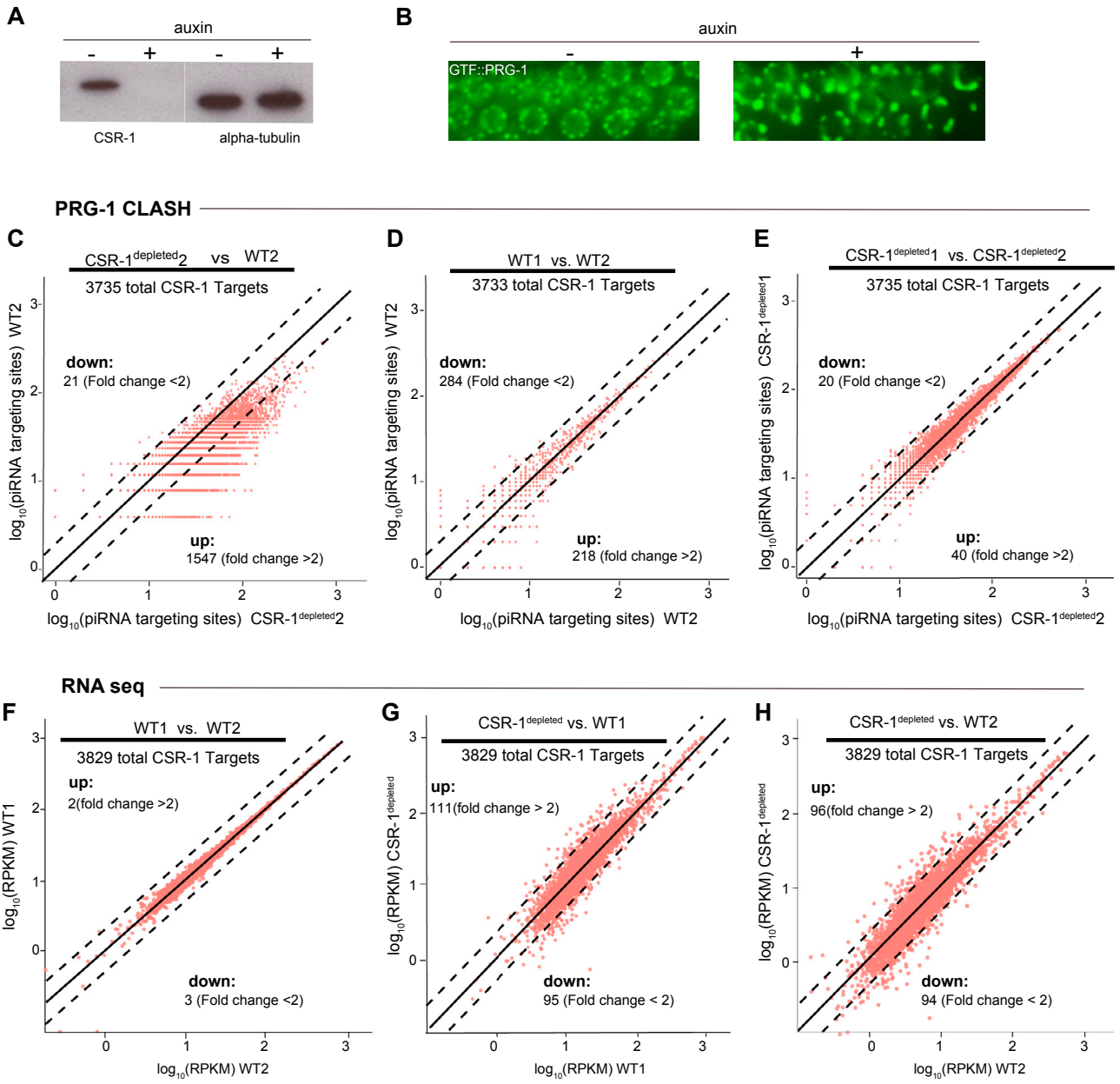


Figure S6. Analysis of CLASH Dataset and mRNA-seq, Related to Figure 6

(A) western blot analysis of the change of protein CSR-1 in the *aid::csr-1* strain without (-) or with (+) auxin treatment. alpha-tubulin was used as loading control.

(B) Localization of GFP::TEV::FLAG::PRG-1 (GTF::PRG-1) in the *aid::csr-1* strain without (-) or with (+) auxin treatment.

(C–E) Comparisons of the number of piRNA target sites in CSR-1 targets between CLASH datasets. Number of piRNA targeting sites in CSR-1 targets that increase or decrease by 2-fold are labeled. CLASH reads were normalized as described in STAR Methods.

(F–H) Comparisons of each CSR-1 targets from RNA-seq datasets. The numbers of CSR-1 targets that increase or decrease by 2-fold are indicated.